

**MACHINE LEARNING-BASED DETECTION AND GIS MAPPING OF THE CROWN- OF
-THORNS STARFISH USING CAMERA-BASED IMAGING**



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BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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CHAPTER 1

INTRODUCTION

Background of the Study

Various biological systems exist on Earth, and coral reefs are among the most varied ones. The ocean is full of life and coral reefs help keep that life going. Coral reefs also help people who live near the ocean by protecting their homes from storms and waves that can wear away the land. Coral reefs are important for people who fish.[1]Coral reefs are in trouble. The people who learn about reefs say that they are getting hurt really badly. One of the problems is the Crown-of-thorns starfish. This is a creature that hurts reefs very much.

The Crown-of-thorns starfish eats an amount of coral every year. To be specific the Crown-of-thorns starfish can eat up to 10 meters of coral each year. The Crown-of-thorns starfish is a problem, for coral reefs. Coral reefs are suffering because of the Crown-of-thorns starfish.Because this feeding behavior occurs continuously, direct destruction threatens the living conditions and health of the coral reefs. If more than 15 starfish per hectare are located in an area, rapid damage is caused to coral reefs because adequate protection is not provided [2].

This study is aligned with Sustainable Development Goal (SDG) 14: Life Below Water, which aims to conserve and sustainably use the oceans, seas, and marine resources. The proposed system uses machine learning and Geographic Information Systems (GIS) to detect, map, and monitor Crown-of-Thorns Starfish (COTS). This helps protect coral reefs, conserve marine biodiversity, and support better marine resource management. By

providing timely detection, mapping, and monitoring of COTS outbreaks, the study supports the protection of marine ecosystems and contributes to coral reef conservation. For people residing in the Philippines, coral reefs provide high value for daily income generation, especially through fish harvesting and travel business. Even though coral reefs have great worth, observation work is restricted because equipment is insufficient and human hand methods are used [4]. Such resource limitations prevent fast tracking and management of COTS outbreaks. If observation work is delayed, greater destruction can fall upon coral reefs.

Old observation methods for COTS rely heavily on underwater swimmers, which requires intense physical labor and remains very slow [5]. Survey activities cover small ocean zones only, particularly in deep or difficult-to-reach locations. Direct human inspection can cause errors, which decreases the correctness of collected data. All these difficulties cause wide-area tracking to become complex and inefficient. This is inefficient and difficult to sustain. There is a need for more efficient and scalable monitoring solutions.

Recent advancements in artificial intelligence have provided more efficient ways to detect the existence of certain species [6]. Convolutional Neural Networks (CNN) and YOLO object detection models are some of the solutions that have been developed for the automated detection of marine species in submerged photographs [5]. These technologies are more efficient and rapid in the detection of species than previous methods used [8]. However, underwater environments present challenges such as poor visibility, light distortion, and water turbidity. To address these issues, image

enhancement techniques are applied to improve image quality and detection performance [9].

Despite these advancements, most existing systems focus only on detection and lack spatial analysis of outbreak locations. There is limited integration of detection systems with Geographic Information Systems (GIS), which are essential for mapping and analyzing environmental data [10]. Plus, a lot of the systems already out there require expensive equipment, making them hard for communities to access. Local areas like the Davao Gulf, as well as Samal and Panabo City, are also having trouble monitoring their COTS populations due to the limited technology that they have access to [11]. This shows the need for a more accessible system that can effectively monitor COTS populations.

The proposed study is aligned with the Department of Science and Technology Harmonized National Research and Development Agenda (DOST-HRNDA) 2022-2028, specifically under the Agriculture, Aquatic, and Natural Resources (AANR) Research and Development Agenda, particularly the Natural Resources and Environment R&D Agenda. This agenda focuses on protecting biodiversity, monitoring ecosystems, and using new technologies to manage natural resources. By using machine learning and Geographic Information Systems (GIS) to detect, map, and monitor Crown-of-Thorns Starfish (COTS), the proposed system supports coral reef conservation and sustainable marine resource management.

The study is also aligned with the Davao del Norte State College (DNSC) Research, Development, and Extension (RDE) Agenda 2026-2030 through the DNSC SAILS Framework. It supports Strategic Priority 1: Sustain

(Sustainability Agenda) by helping protect biodiversity, preserve marine ecosystems, and support sustainable environmental practices. It also supports Strategic Priority 3: Innovation and Inclusion (I² Agenda) through Information and Communication Technology for Development (ICT4D) by using machine learning and Geographic Information Systems (GIS) to develop an innovative system for monitoring Crown-of-Thorns Starfish (COTS) and protecting coral reefs.

Furthermore, the study is aligned with the Commission on Higher Education (CHED) A.C.H.I.E.V.E. Agenda 2025-2030, particularly the H - Harmonized SDG-Based Research and Innovation thrust by supporting research that contributes to Sustainable Development Goal (SDG) 14: Life Below Water. It also supports the I - Integrated Real-Time Data Collection and Analytics System enabler by using Geographic Information Systems (GIS) and machine learning to collect, analyze, and display data that can help in making better decisions for coral reef conservation and marine resource management.

This study is trying to solve some problems. It wants to create a system that uses machine learning and maps to find Crown-of-thorns starfish in the water. The system will use a kind of map called Geographic Information System to do this. It will look at pictures taken by cameras to find the starfish. The people working on this project want to make it easy to use. They are going to try using GoPro cameras to take pictures underwater. These cameras can take good pictures even in the water. This will help them put the system in the ocean where it can do its job.

The main goal of the system is to find the Crown-of-thorns starfish on its own show where they are, on a map and count how many there are. This will help people who take care of the reefs make good decisions and keep the reefs safe. The system will use Crown-of-thorns starfish pictures to do all of this and help with coral reef conservation efforts. Additionally, the system will include a field contribution module that allows authorized field participants such as divers, marine personnel, and conservation volunteers to submit Crown-of-Thorns Starfish (COTS) removal records and ecological monitoring information. This enhancement supports postdetection conservation monitoring by documenting removal activities, removal quantity, and field observations to strengthen coral reef management efforts.

Objectives of the Study

The main objective of this project is to develop a machine learning-based system integrated with GIS for detecting and mapping Crown-of-thorns starfish (COTS) using camera-based imaging. Specifically, the study aims to:

1. To develop and compare the performance of selected object detection models, namely YOLOv8, YOLOv11, YOLO26, for detecting Crown-of-Thorns Starfish (COTS) in underwater images, and determine the best-performing model using accuracy, precision, recall, F1-score, and detection speed metrics.
2. To develop a GoPro-assisted detection system integrated with a mobile/web platform capable of processing uploaded underwater images and identifying Crown-of-Thorns Starfish (COTS).

3. To develop a Geographic Information System (GIS) that maps detected COTS locations, automatically counts detected individuals, and visualizes their spatial distribution.
4. To develop intervention planning and monitoring features that support the management of COTS-affected reef areas through the recording of removal activities, status updates, and field monitoring information.
5. To generate analytics, visualizations, and summary reports that support marine ecosystem monitoring, outbreak assessment, and conservation decision-making.

Significance of the Study

This study introduces an AI-driven and GIS-integrated approach to marine ecosystem monitoring by transforming traditional manual observation into an intelligent and spatially aware detection system. The system contributes to improved monitoring efficiency, spatial analysis, and timely environmental decision-making for coral reef conservation. The following are the beneficiaries of the study:

For Marine Conservation and Environmental Monitoring Sectors. This system is really good for people who want to take care of the environment. It helps us find out where the Crown-of-Thorns Starfish are. The Crown-of-Thorns Starfish can hurt reefs. So this system is very useful. It can automatically find the Crown-of-Thorns Starfish. Show us where they are. This way people can do something about it before it is too late. The coral reefs get hurt. The Crown-of-Thorns Starfish are a problem, for coral reefs. Plus, by integrating field data with conservation reporting, reef management strategies and ecological monitoring can be improved.

For Environmental Researchers and Marine Scientists. Utilize the integration of machine learning, underwater imagery, and Geographic Information Systems (GIS) to carry out marine spatial analysis and ecological monitoring. Furthermore, this study may open doors for the future development of AI-based marine biodiversity assessment tools, marine ecosystem monitoring, and reef conservation technologies.

Local Government Units (LGUs) and Coastal Resource Managers. The mapped spatial data generated by the system may support coastal resource planning, marine protected area management, and environmental policy formulation. Visualizing where outbreaks of certain reef-destroying organisms occur can help in the planning and management of reef conservation programs and aid.

Fisherfolk and Coastal Communities. Rely on the fish populations of these reefs. By improving reef monitoring and enabling earlier detection of COTS outbreaks, the system may contribute to the long-term sustainability of fisheries and help to reduce the ecological damage that affects the fish populations and the livelihoods of coastal communities. By maintaining the health of the reef ecosystems, the system may also help to sustain the livelihoods of those coastal communities.

Tourism and Marine Ecosystem Protection. Experts say that coral reefs are really important because they are nice to look at and people like to dive and snorkel near them. When people who work to protect the environment control the starfish that can hurt reefs it helps to keep the coral reefs from getting damaged. This is good for reefs because it helps keep them looking

nice which is good for people who come to visit and it also helps the people who live near the coral reefs.

Students and Future Researchers. This study can provide a reference for those who wish to pursue further studies in the various fields that are touched upon in this paper. Additionally, this study has the potential to lead to the development of new and improved systems for managing and protecting the marine environment.

Scope and Limitation

This study is about creating and testing a system that uses intelligence and Geographic Information System technology to find and record Crown-of-Thorns Starfish from pictures taken underwater. The system is meant to be used in the reefs of Davao Gulf, especially close to Samal and Panabo City. It uses pictures taken mainly with GoPro cameras that are made for looking at areas, under water. The GoPro device supports underwater image acquisition at varying depths, improving field data reliability.

The system supports static underwater image uploads captured during reef monitoring activities. Detection and analysis are performed using uploaded underwater images processed by the machine learning model. This approach enables efficient and cost-effective field data collection while maintaining image quality for marine monitoring purposes.

The system operates in an image processing environment rather than continuous live monitoring. Detection and GIS analysis are performed after user upload captured underwater images through the mobile application. Processing performance depends on hardware capability, such as mid-range smartphones or laptops with moderate specifications. Under normal operating

conditions, machine learning inference and GIS visualization complete efficiently for uploaded image batches, with performance varying according to image resolution and system workload. All processing is performed through cloud-based inference integrated with the mobile application.

For geographic mapping, the system uses mobile device GPS coordinates obtained from the connected mobile application and synchronized with GoProcaptured images during field submission. GPS coordinates are automatically associated with uploaded monitoring data without requiring manual coordinate entry or external GPS hardware. GPS accuracy may vary due to environmental conditions and satellite signal availability in coastal monitoring areas, which may result in minor positional variation in GIS visualization outputs.

The system covers image acquisition, preprocessing, machine learning-based object detection, GIS mapping, automatic counting of Crown-of-Thorns Starfish, and field contribution monitoring features. In addition to automated detection and mapping, the system includes a field contribution module that allows authorized field participants, including divers, marine personnel, and conservation volunteers, to submit Crown-of-Thorns Starfish (COTS) removal records and ecological monitoring information. Submitted records may include removal quantity, location information, and supporting field observations related to reef monitoring activities. System outputs include detection results, mapped spatial distribution, monitoring records, removal monitoring information, and analytical summaries to support marine monitoring and conservation decision-making. The system is intended for practical reef monitoring activities and marine conservation support.

This study is limited to image-based detection, automatic counting, spatial visualization of COTS populations, and field-contributed monitoring information. Submitted monitoring records depend on authorized participant reporting and validation procedures to ensure consistency and reliability. The system does not include physical or automated removal mechanisms, as removal activities remain under authorized field personnel. Environmental conditions such as turbidity, lighting variation, underwater visibility, and image quality may affect detection performance. The system also does not include predictive modeling of starfish movement based on oceanographic or environmental factors due to the need for larger datasets and more advanced computational modeling, which are recommended for future work.

Review of Related Literature

This section reviews existing literature related to machine learning, computer vision, underwater imaging, and Geographic Information Systems (GIS) in marine ecosystem monitoring. It examines current technologies and methodologies relevant to the detection and analysis of Crown-of-thorns starfish (COTS) and provides the foundation for the development of the proposed detection and mapping system.

The advancement of artificial intelligence (AI) and computer vision technologies has significantly improved marine ecosystem monitoring. Deep learning approaches such as Convolutional Neural Networks (CNNs) and YOLO-based object detection models are widely used for marine species identification and ecological image analysis [12]–[17]. Studies have shown that these approaches improve automated detection accuracy, scalability, and monitoring efficiency under varying underwater conditions. These models are

particularly effective in identifying marine organisms from complex underwater backgrounds where manual observation may be inconsistent. Data augmentation and preprocessing techniques further improve model robustness and detection reliability in complex marine environments [5], [18], [19]. Traditional monitoring methods for Crown-of-thorns starfish rely heavily on diver-based surveys, which are labor-intensive, time-consuming, and limited in spatial coverage [2]. In contrast, AI-assisted monitoring systems and underwater imaging technologies provide faster and more scalable monitoring solutions [24], [25], [29], [30]. Underwater image processing techniques such as contrast enhancement, dehazing, and color correction are commonly applied to improve image quality and detection reliability under varying marine conditions [8], [20], [21].

Recent developments in underwater imaging technologies have improved marine ecosystem monitoring through the use of compact and high-resolution camera systems. Underwater photogrammetry and diver-operated imaging systems provide detailed visual data suitable for ecological observation and automated analysis [32], [33]. Portable underwater cameras such as GoPro devices are practical for field deployment because of their waterproof capability, high-resolution imaging, compact design, and compatibility with mobile systems [32], [36]. Geographic Information Systems play a part in studying the ocean and the space it takes up making maps of reefs and showing how the environment looks [9], [22], [23]. Geographic Information Systems help put and show pictures of information, about the environment so researchers can use Geographic Information Systems to understand things better to identify affected reef zones and monitor

biodiversity distribution. Recent GIS developments incorporate spatial modeling, heat mapping, and interactive dashboards to improve marine conservation planning and environmental decisionmaking [26], [27]. The Geographic Information System visualization is really helpful for people who are doing research and for groups that take care of the environment. It helps them find the parts of the reef that're in big trouble and decide what to do first to help.

In the Philippines people who take care of the ocean are using computers and Geographic Information System technology more to watch over the coral reefs and make sure they are safe. This is good for coral reef conservation, in the Philippines. The Bureau of Fisheries and Aquatic Resources (BFAR) [4] reports that coral reef ecosystems face threats from destructive fishing practices and biological stressors such as COTS outbreaks. Studies from the University of the Philippines Marine Science Institute also show that more and more people are using geospatial technologies and AI-based systems to monitor reefs for coastal resource management [10].

Managing outbreaks of Crown-of-Thorns Starfish requires not finding them but also acting quickly and keeping track over time to stop coral reefs from getting worse. The Great Barrier Reef Marine Park Authority says that the best way to handle this is to use a plan that includes finding the starfish early checking how bad the problem is removing them fast in the right places and checking the areas again over time to stop the problem from getting worse. These ways of working help people in charge of the ocean to focus on the areas that're most at risk keep records of what they do and see if their

actions are working through regular checking. This kind of work helps make choices and makes it easier to protect coral reefs in the long run.

This way of working is what the COTS Monitor: Crown-of-Thorns Starfish Detection and GIS Mapping System is based on. The system uses machine learning to find the starfish makes maps with GIS gets help from experts sends alerts and suggests what to do. Of managing the actual actions the system helps people make decisions by finding the areas that are most at risk suggesting what to do keeping records of what is done and helping to check again later. This way helps people, in charge of the ocean to focus on the important actions and keep good records for future planning.

Recent environmental monitoring approaches also incorporate participatory ecological monitoring strategies to strengthen conservation initiatives and long-term environmental assessment [37], [38]. Community-assisted monitoring allows trained field participants to contribute ecological observations that complement automated AI and GIS-based systems. The integration of field-contributed monitoring information with automated analytical technologies supports broader spatial coverage and improved conservation assessment [38], [39]. Field-contributed monitoring information also supports validation of automated detection results and improves conservation response documentation.

Despite these advancements, several gaps remain evident in existing studies. Most research focuses primarily on detection and classification tasks, with limited integration of GIS-based mapping and spatial visualization systems. Additionally, many existing solutions rely on expensive equipment and controlled research environments, limiting accessibility for local and

community-based applications. These limits show that we need a system that works together is easy to use and uses GIS to find and map things.

The papers we looked at say that AI and GIS help monitoring a lot. They make it better, at finding things help with looking at where thingsre make it easier to see the environment. However, the lack of unified systems that combine detection, counting, GIS-based mapping, and fieldcontributed monitoring information continues to limit real-world applicability. Therefore, this study aims to develop an integrated system for the detection, counting, and GISbased mapping of Crown-of-thorns starfish to support improved marine ecosystem monitoring and conservation decision-making.

Related Works

Several systems have been developed for marine species detection and environmental monitoring using artificial intelligence and underwater imaging technologies. These systems commonly focus on object detection, classification, and monitoring of marine organisms in underwater environments. Many systems utilize CNN-based and YOLO-based models for automated detection and counting, achieving high detection accuracy and monitoring efficiency. Some studies also incorporate Geographic Information Systems (GIS) for reef mapping and spatial visualization. However, many existing systems still lack integrated mobile-based deployment, field contribution monitoring, and accessible GIS-enhanced monitoring capabilities for local conservation applications. These limitations highlight the need for a more integrated and accessible monitoring system. Compared to existing systems, the proposed study introduces a more comprehensive and integrated approach to marine ecosystem monitoring. While most systems

focus only on detection and classification, the proposed system incorporates Geographic Information System (GIS) for spatial mapping of detected species. The system also utilizes GoPro-based underwater imaging, making it more accessible and cost-effective compared to systems that rely on specialized equipment. Although some studies explore predictive analytics for estimating movement patterns, such approaches require complex modeling and real-time environmental data. As a result, the proposed system provides a more practical, scalable, and accessible solution for monitoring Crown-of-thorns starfish population.

System	Detection Accuracy	GIS Mapping	Monitoring Support	Mobile-Based Deployment	Automated Counting	Field Contribution
CSIRO COTS Detection Dataset	✓					
CNN-Based Fish Detection			✓			
Diver-Based Manual Survey			✓			✓
GIS-Based Coral Reef Mapping		✓	✓			
Deep Learning for Fish Classification	✓		✓			
Proposed System	✓	✓	✓	✓	✓	✓

Table 1. Comparison table of existing system vs. the proposed system.

The CSIRO Crown-of-Thorn Starfish Detection Dataset

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 Andy Steven^{*} Nic Heaney[†] Karl von Richter[†] Lachlan Tychsen-Smith[†]
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Figure 1: A COTS outbreak at the Swain Reefs, in North Queensland, Australia, October 2021.

ABSTRACT

Crown-of-Thorn Starfish (COTS) outbreaks are a major cause of coral loss on the Great Barrier Reef (GBR) and substantial surveillance and control programs are underway in an attempt to manage COTS populations to ecologically sustainable levels. We release a large-scale, annotated underwater image dataset from a COTS outbreak area on the GBR, to encourage research on Machine Learning and AI-driven technologies to improve the detection, monitoring,

1 INTRODUCTION

Australia's Great Barrier Reef (GBR) is a national icon and a World Heritage Site, where tiny corals build continental-scale underwater structures inhabited by thousands of other marine creatures. While it is no secret that the GBR is under threat, it may not occur to everyone that a species of starfish, the Crown-of-Thorns Starfish (COTS), is one of the few main factors responsible for coral loss on the GBR.

Figure 1. CSIRO Crown-of-Thorns Starfish Detection Dataset

The CSIRO Crown-of-Thorns Starfish Detection Dataset is a group of pictures taken underwater that have been marked up. These pictures are used to teach and check how well AI models can find Crown-of-Thorns Starfish. The main goal is to make finding these starfish more accurate by giving researchers data to work with.. This dataset is only for training AI models. It does not have things like maps tracking help moving tools, counting or help from the field. So it is good for making AI models better but not, for a system to watch the ocean.

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FIGURE 1 Illustration of four typical types of CV tasks From left: Image Classification (*i.e.* is there a fish in the image, or what type (class) of fish is in the image?), Object Detection/Localisation, Semantic Segmentation, Instance Segmentation.

a class of DNN, most commonly applied to visual analyses. For instance, CNNs have been successfully used for analysis of fish habitats (Xu et al., 2019; Konovalov et al., 2019a; Pope et al., 2010). In comparison to other image recognition algorithms, CNNs have the significant benefit that they require limited pre-processing. CNNs are not hand-engineered but uncover and learn hidden features in the data on their own. They learn level-by-level with various levels of abstraction.

underwater fish habitat monitoring. In addition, we provide insights into challenges and opportunities in the underwater fish habitat monitoring domain. It is worth noting that our article is written to provide a general and high-level, as opposed to detailed, introduction of deep learning and its relevant contexts for marine scientists. This is useful in understanding the follow-up discussions on the use of deep learning in the marine task of underwater fish classification.

Figure 2. Fish Detection System

The fish detection system based on CNN uses learning to find and recognize different types of fish in underwater pictures. The main job of this system is to make detection more accurate and help with keeping track of the ocean life by looking at underwater information. This makes it easier for scientists to watch fish populations and saves time that would otherwise be spent looking at pictures.. The system does not include things like GIS maps, a version that works on phones, automatic counting or help, for people working outside in the field because it is mainly made for finding and watching fish.

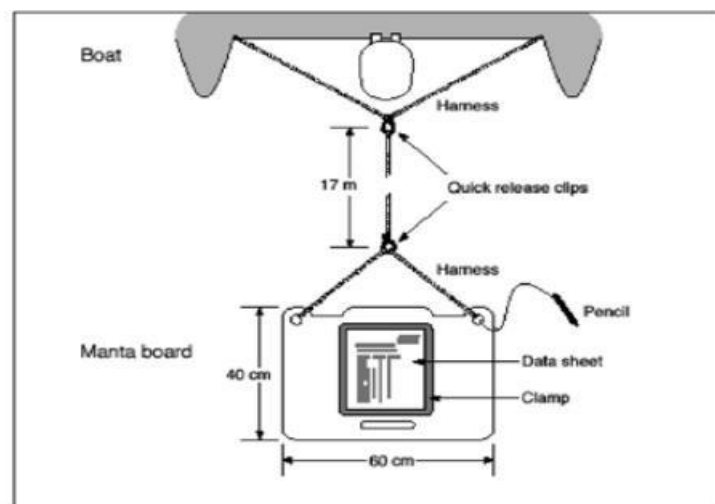


Figure 3. AIMS Diver-Based Manual Survey

The AIMS Diver-Based Manual Survey is a way to check how healthy coral reefs are. This method uses SCUBA divers and manta tow surveys to look for Crown-of-Thorns Starfish. The main purpose is to help keep track of the reefs by letting trained divers see, count and write down how Crown-of-Thorns Starfish there are. They also record how much coral is there and any damage, to the coral. The information gathered helps scientists understand how the reefs are doing. This information also helps people take care of the

reefs and manage them better. This survey does not use GIS mapping. It does not use phones to help with surveys. It does not use machines to count things. It does not use computers to find things. All the work is done by people who're underwater.

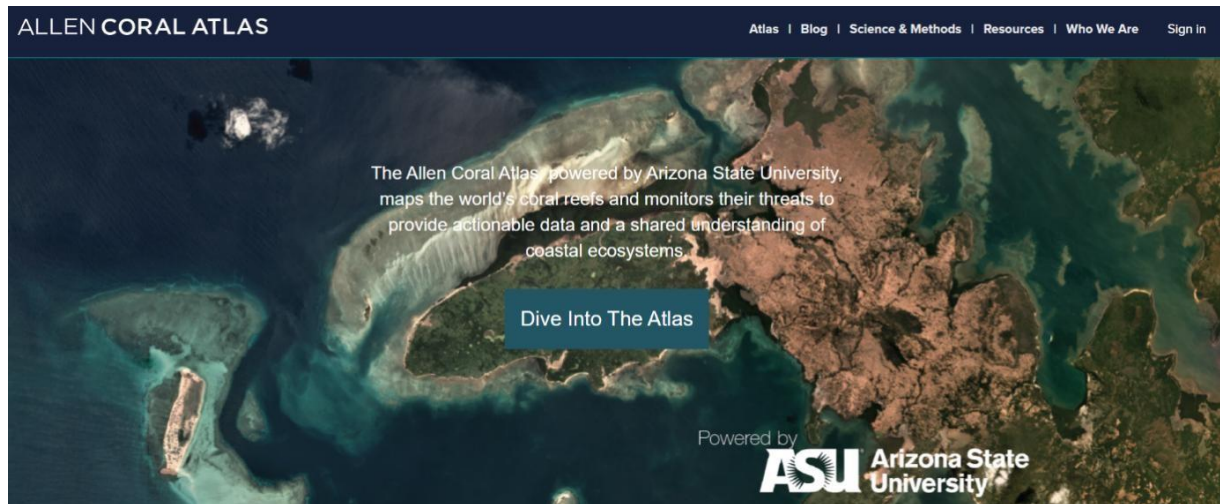


Figure 3. Allen Coral Atlas

The Allen Coral Atlas is a tool that uses pictures from space and special computer programs to make maps of reefs all around the world. It helps people who study the environment and work to protect it by giving them maps they can use on their computers. These maps show what is going on with the reefs and help people figure out which areas need the most help. The Allen Coral Atlas is good at showing where the coral reefs are and what they look like. The Allen Coral Atlas provides maps that show where the coral reefs are and what is happening to them. However the Allen Coral Atlas does not have some features that might be useful such as being able to find Crown-of-Thorns Starfish or being able to use it on a phone or being able to count things automatically or letting people add information from the field. The Allen Coral Atlas is still an useful tool, for people who care about coral reefs.

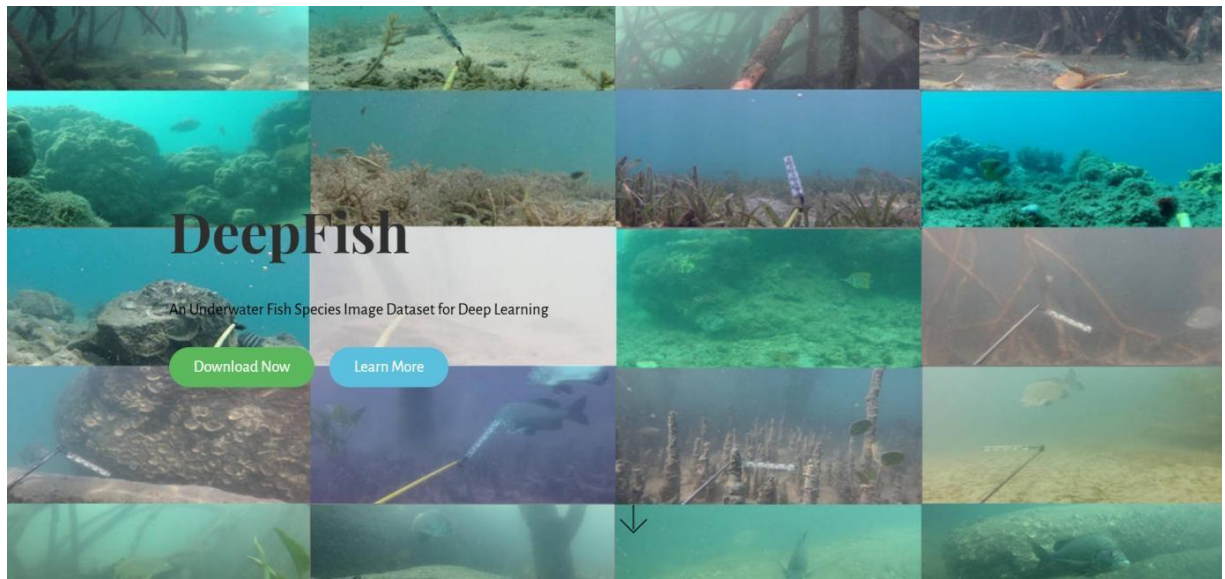


Figure 4. DeepFish

DeepFish is a system that uses learning to classify fish. It looks at pictures and videos from underwater. Figures out what kind of fish are in them. The main thing DeepFish does is help people detect fish accurately. This is good for people who study the ocean because it helps them keep an eye on things without having to do it all by hand. DeepFish helps researchers find out what kind of fish are in the water and look at what's happening underwater more easily than if they had to do it themselves. DeepFish is for looking at fish and figuring out what kind they are. It does not do things like make maps or work on phones. It also does not count fish automatically. Let people add information from the field. DeepFish is only, about finding and classifying fish.

Definition of Terms

Artificial Intelligence (AI). Refers to computational techniques used to detect anomalies and identify Crown-of-thorns starfish (COTS) from underwater images.

Detection. The process by which the AI model identifies instances of Crown-of-Thorns Starfish (COTS) in images and automatically counts each detected instance.

Crown-of-thorns starfish (COTS). A coral-eating predator (*Acanthaster planci*) capable of consuming up to 10 m² of coral annually, posing a major threat to reef ecosystems.

Davao Gulf. A major marine ecosystem in the Philippines, including Samal and Panabo City, serving as the local study site for this project.

Geographic Information System (GIS). A system designed to capture, store, analyze, and visualize spatial data, used in this study to map COTS outbreaks.

GoPro-Based Imaging. The use of GoPro underwater cameras for capturing high-resolution images for COTS detection.

Image Preprocessing. Techniques such as contrast adjustment and color correction applied to underwater images to improve clarity and detection performance.

Outbreak. A population of Crown-of-thorns starfish exceeding 15 individuals per hectare, considered destructive to coral reefs.

Removal Record. Structured monitoring information documenting Crown-of-Thorns Starfish removal activities including removal quantity, location, observations, and validation status.

Underwater Imaging. The process of capturing marine visuals using GoPro underwater cameras and waterproof imaging devices for ecological monitoring.

You Only Look Once (YOLO). A real-time object detection algorithm capable of identifying multiple objects in a single image frame.

CHAPTER 2

METHODOLOGY

The project adopts a Hybrid Project Management Methodology that combines the structured approach of Waterfall and the flexibility of Agile to effectively support the requirements of the study, as illustrated in Figure 1. The Waterfall approach is applied during the initial stages, particularly in planning, requirements definition, and dataset preparation, ensuring that all outputs such as system design, data collection strategy, and model requirements are clearly defined before development begins. Meanwhile Agile methods are used during the implementation and testing parts of the project. This allows for steps of development, regular checking and small improvements to system features. These features include things like image-based detection, GIS mapping and automatic counting. This way there is a mix, between a clear plan and being able to change as needed. This approach works well for systems that have steps and also parts that change over time. These parts include machine learning models and mobile app designs.

The way the project is done follows the framework. This framework has five parts: Planning, Analysis, Design, Implementation and Maintenance.. Because machine learning systems change a lot Agile sprint cycles are added during the implementation and testing parts. These sprints help to keep improving the model accuracy how easy it is to use. The GIS features. This combination allows the project to maintain structured documentation while remaining adaptable to changes encountered during model training and underwater field testing.

To support incremental development and continuous refinement of the proposed system, the study incorporates Agile sprint cycles within the implementation and testing phases. Each sprint cycle focuses on a specific component of the system and allows the proponents to gradually improve the machine learning model, mobile application, GIS mapping functionality, and database integration through iterative development and evaluation.

The project is split into a parts. We have the part that is all, about getting a lot of underwater pictures and getting them ready to use. We get these pictures we organize them and we put labels on them so we can find the Crown-of-thorns starfish, also called COTS easily. In this part we make sure the pictures are good to use for our project. We do things like make the pictures clearer we check the labels again to make sure they are correct we make sure we have the number of pictures and we check everything one more time to make sure our data is good and we can trust it for the Crown-of-thorns starfish project.

The third sprint focuses on initial model training and detection development using the YOLO-based object detection algorithm. During this sprint, the model is evaluated using metrics such as accuracy, precision, recall, and F1-score. The fourth sprint involves integrating the trained model into the mobile application and GIS mapping system, enabling spatial visualization of detected COTS locations using GPS coordinates and reef zone information. The fifth sprint focuses on system testing and optimization, where functional testing, usability testing, and detection validation are conducted to identify errors and improve system performance. The last sprint includes checking the system improving the documents and getting ready, for

putting it into use and showing it off. During these repeated sprint cycles the people who started the project keep making the system better. to improve detection accuracy, mapping reliability, usability, and overall system performance.

The Planning phase is the step in the PADIM framework. It sets the direction for the project by looking at its goals what it will cover, when it will happen what resources are needed and what results are expected. This step is the base of the project. Gives a clear plan for the next steps, in building the system. In the Analysis phase , focuses on understanding and identifying the requirements of the proposed system. This is the part where we figure out what the system needs to do and what it should be able to handle. We are talking about the system and the system has to meet certain requirements. The system requirements are very important. The system has to support things like information and processes and what the users of the system need. We need to make sure we understand what the system requirements are before we start designing the system. This is a step, for the system.

In the Design phase is where the overall structure and blueprint of the system are prepared. It describes how the system will be organized, including its architecture, database, user interface, and other major components. This phase provides a clear guide for developing the proposed system. The implementation phase is the stage where the approved system design is transformed into a working system. During this phase the different system components are developed, integrated and tested to make sure that they work properly and meet the needed specifications. Since the study uses a

SYSTEM PLANNING

Project Team Organization

The project team, as shown in Figure 2, is organized to ensure effective coordination and clear distribution of responsibilities in the development of the Crownof-thorns starfish detection and mapping system. The Adviser provides overall guidance and ensures that the project meets academic, technical, and research standards, particularly in the application of machine learning and GIS technologies.

The Project Manager oversees planning, task coordination, dataset preparation activities, and progress monitoring to ensure that all phases of development are completed on time. The people who help make this project work include the Machine Learning Developer. The Machine Learning Developer is in charge of collecting datasets for the Machine Learning Developer. They also have to annotate these datasets for the Machine Learning Developer. Then they train models for the Machine Learning Developer. Make sure the detection algorithm for the Machine Learning Developer is working as well as it can.

There is also the Mobile and GIS Developer. The Mobile and GIS Developer does a few things. The Mobile and GIS Developer integrates systems for the project. The Mobile and GIS Developer also develops applications for the project. The Mobile and GIS Developer is responsible for mapping functionalities, for the project. The person who works as a System Analyst and also handles Documentation Specialist tasks is, in charge of understanding what is needed for a project creating system design documents and making sure everything that comes out matches the goals of the project

and what is planned. This role helps everyone work together well and manages both the side and the documentation side of the project in a balanced way.



Figure 6. Project Management Team Organization

Work Breakdown Structure

The Work Breakdown Structure is a way to break down the project into parts. This is shown in Figure 3. It follows the framework. The PADIM framework is used to organize the project into stages. These stages are Planning, Analysis, Design, Implementation and Maintenance. The Work Breakdown Structure helps us manage the project. It makes sure we do all the tasks in the order. We have to do things to make the machine learning-based detection and GIS mapping system for Crown-of-thorns starfish work.

We have to collect data and label it. We have to train the model. We have to design the system. We have to make an application. We have to test it and see if it works. The Work Breakdown Structure is like a plan. It helps us stay organized. We can see what needs to be done and when. This makes it easier to manage our time and make sure everything gets done. The Work Breakdown Structure is very helpful for the machine learning-based detection and GIS mapping system, for Crown-of-thorns starfish project. It helps us do all the tasks and write all the documents we need.

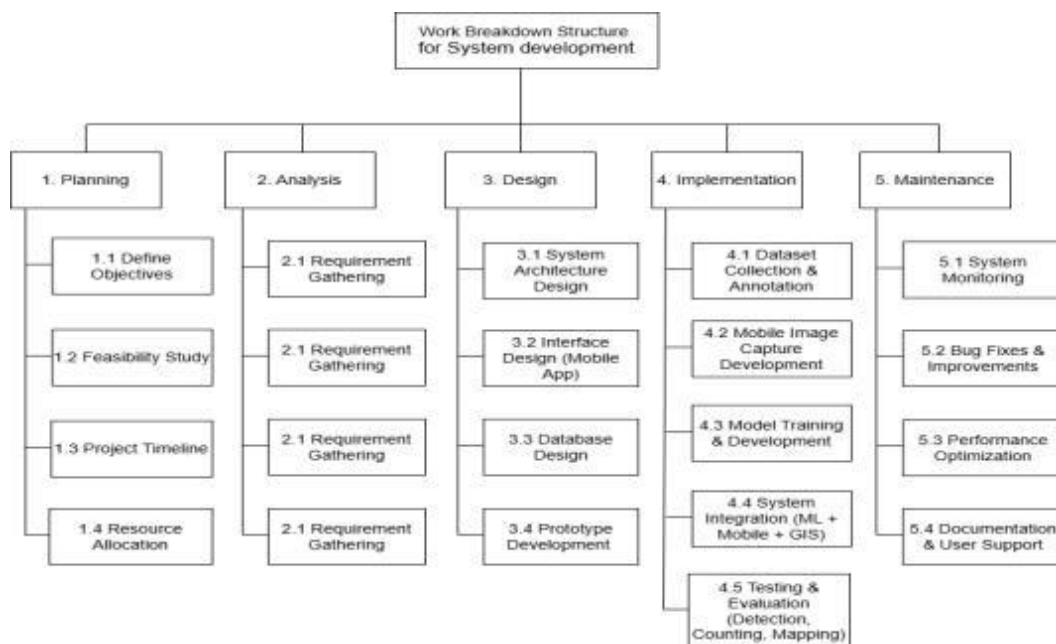


Figure 7. Work Breakdown Structure.

Gantt Chart

The Gantt chart in Figure 4 presents the timeline and schedule of activities for the development of the Crown-of-thorns starfish (COTS) detection and GIS mapping system, aligning all tasks with the PADIM framework. The sequence of the project is shown in a chart. This chart shows how long each part of the project will take. It starts with Planning and Analysis. Then goes to Design, Implementation and Maintenance. All the work that needs to be done is included, like figuring out what is needed getting the data

ready designing the system making a model putting it on an application testing it and writing it all down. The chart makes sure all this work is done on time. It is easy to see how each part of the project is connected and how they can be done at the time if needed. This helps to save time. The people in charge can use the chart to keep track of what needs to be done and when. They can make sure important things like training the model and putting the system together are done well. The Gantt chart helps with managing time tracking progress and finishing the capstone project on time. The Gantt chart is very helpful, for the capstone project. It supports the capstone project by making sure everything is done when it should be.

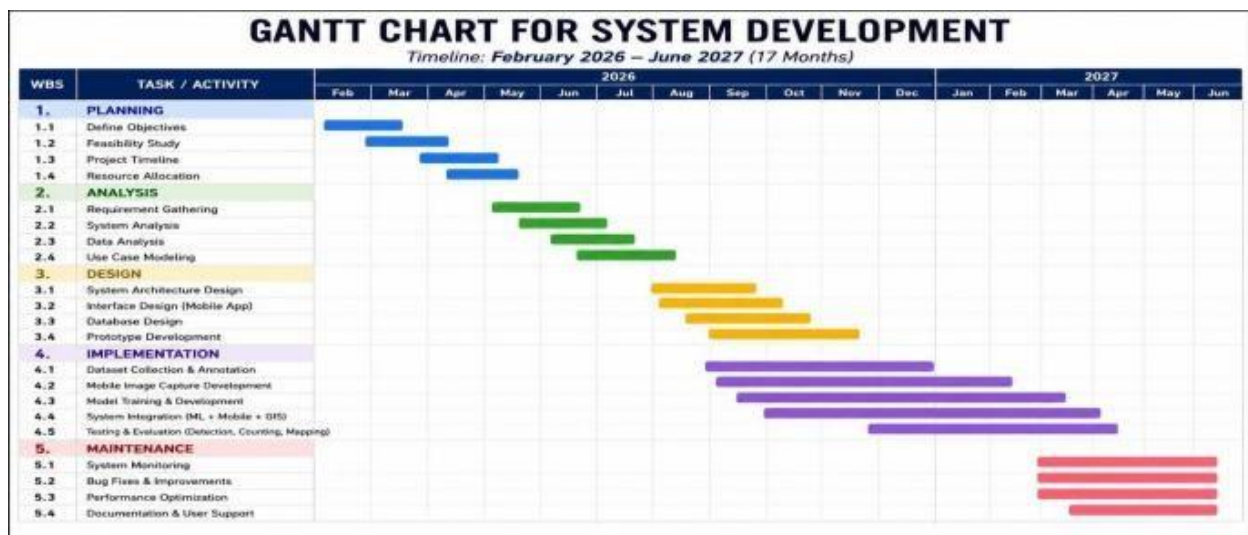


Figure 8. Gantt Chart of the system

SYSTEM ANALYSIS

System Architecture

The system architecture of COTS Monitor integrates a mobile application, AI detection model, GIS mapping, admin dashboard, and centralized database to support continuous monitoring and conservation activities. Field Officers or Divers use GoPro cameras to capture underwater images of reef environments and upload them to the mobile application. The

application serves as the primary platform for image submission, monitoring, and field data collection. The mobile application also supports batch uploading, allowing Field Officers or Divers to upload multiple GoPro images during a single monitoring activity instead of one image at a time. To ensure accurate spatial mapping, images are uploaded while the diver is within the target reef zone, enabling the system to associate each submission with the corresponding GPS coordinates. It also records environmental and geographical information such as GPS coordinates, date, time, reef zone, and estimated depth. These automated features improve monitoring accuracy and reduce manual data entry.

After image submission, the system applies preprocessing techniques such as color correction, contrast enhancement, and image quality improvement to address underwater visibility challenges. The processed images are then analyzed using a YOLO-based AI detection model to identify and count possible Crown-of-Thorns Starfish. The system generates confidence scores to measure the reliability of detection results. If the confidence level is low, users may be asked to retake or upload clearer images. Validated detections are stored in the centralized database together with all related monitoring information. Once a Crown-of-Thorns Starfish (COTS) is successfully detected, the system automatically plots the detection location on the GIS map using a red marker based on the recorded GPS coordinates. This enables administrators to immediately visualize the spatial distribution of detected COTS, monitor affected reef zones, and support timely conservation and intervention planning.

The mobile application also supports conservation and field reporting activities by allowing authorized users to submit removal records, ecological observations, reef condition reports, and other monitoring data. Secure login authentication ensures that only authorized personnel can access system functions. These field contributions provide additional information that supports marine conservation efforts. All submitted records are linked to corresponding monitoring data and locations. This creates a more comprehensive and reliable monitoring database.

Administrators manage the system through a web dashboard equipped with modules for GIS monitoring, records, alerts, analytics, reports, users, and settings. Through these tools, they can validate detections, monitor affected reef zones, analyze COTS population trends, and generate reports for conservation planning. The system also provides automated alerts for high-risk areas experiencing increased COTS activity. All data are stored in a centralized database that supports continuous monitoring, historical analysis, and decision-making. Overall, the integration of AI, GIS, mobile technology, and underwater imaging provides an efficient and reliable solution for monitoring Crown-of-Thorns Starfish outbreaks in marine ecosystems.

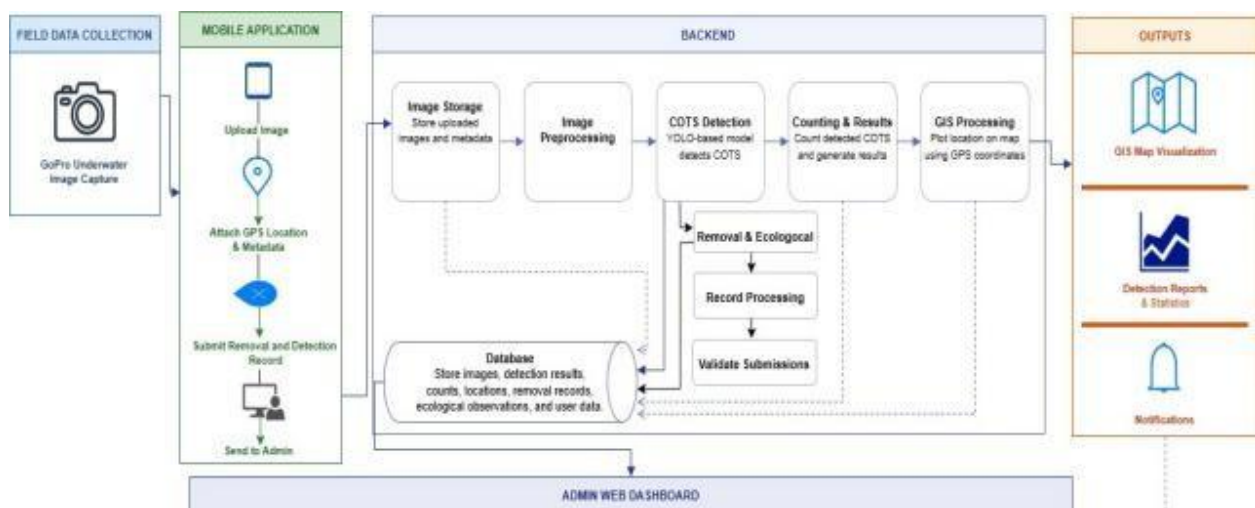


Figure 9. System architecture of the project.

Conceptual Framework

The conceptual framework of this study follows the Input–Process–Output (IPO) Model, which illustrates how the proposed machine learning-based system transforms raw underwater monitoring data into meaningful detection, mapping, and conservation outputs. The framework demonstrates the relationship among the required inputs, the processing stages performed by the system, and the expected outputs that support marine ecosystem monitoring and decision-making. The IPO model serves as a guide in understanding the overall workflow

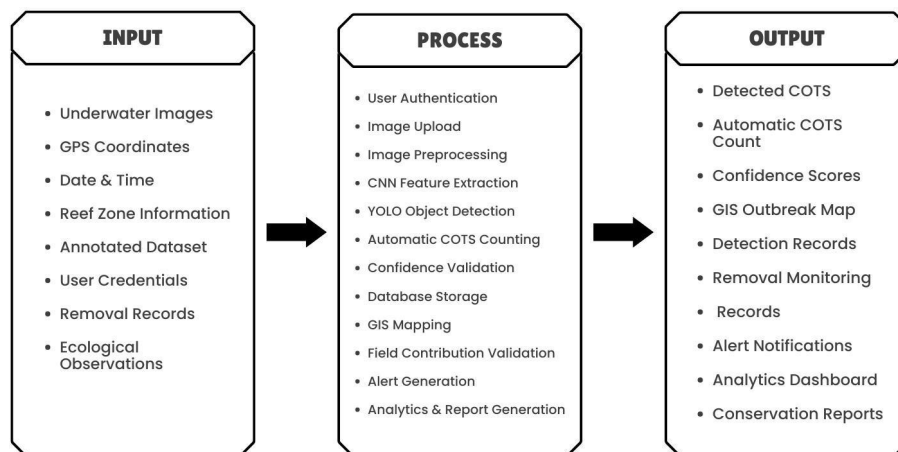


Figure 10. Conceptual Framework.

This conceptual framework is aligned with the proposed system architecture of the study. It illustrates how underwater images, GPS information, and field monitoring data are used as the primary inputs of the system. These inputs go through a lot of steps like getting images using machine learning to find things working with geographic information systems and managing databases to get good and trustworthy information, about what is happening. The information that has been processed is then changed into

things like results that show where COTS are maps that use information systems, automatic warnings reports that analyze things and records of conservation efforts related to COTS and conservation. Overall, the conceptual framework demonstrates the relationship among the major components of the study and explains how the proposed system supports the detection, mapping, monitoring, and conservation of Crown-of-Thorns Starfish populations.

Functional and Non-functional Requirements

The functional and non-functional requirements define the expected features and quality attributes of the COTS Monitor: Crown-of-Thorns Starfish Monitoring System. The functional requirements describe the main operations the system must perform, such as detection, monitoring, alert management, report generation, and user management. The non-functional requirements describe how well the system should work in terms of how fast it is, how reliable it is, how easy it is to use how secure it is, how well it works with other systems how easy it is to maintain and how available it is. These non-functional requirements are important because they help make sure the system supports the people who use it every day and the people who manage it. They are listed in a list and they follow the quality attributes that are, in the ISO 25010 which is what the methodology guide says we have to do. The system shall provide the following functionalities:

1. The system shall allow authorized users, including Field Officers, Divers, and Administrators, to log in securely.
2. The system shall allow Field Officers or Divers to capture underwater image using GoPro cameras and upload them to the system.

- 3.The system shall process submitted images and display detection results, including COTS count, confidence level, date, time, GPS location, and zone information.
- 4.The system shall validate detection results and reject or require review when the confidence level is low.
- 5.The system shall allow valid detection records to be submitted and stored in the centralized database.
- 6.The system shall allow Administrators to monitor reef zones, view submitted records, and see AI-assisted zone status suggestions (Active, Scheduled, Cleared).
- 7.The system shall generate alerts when submitted detections indicate possible COTS infestation or high-risk activity.
- 8.The system shall allow Administrators to review, manage, and update alert status.
- 9.The system shall provide map monitoring to display affected zones, monitored areas, and COTS locations.
- 10.The system shall provide analytics and generate reports based on detection records, zone activity, alerts, and monitoring data.
- 11.The system shall allow Administrators to manage user accounts, roles, permissions, and system settings.
- 12.The system shall allow authorized field participants to submit COTS removal records.
- 13.The system shall store removal quantity information.
- 14.The system shall allow submission of ecological monitoring observations.

15. The system shall display recommended intervention measures for reef zones classified as High infestation after expert validation.

The following are the system's non-functional requirements:

1. Performance Efficiency. The system shall process image submissions and display detection results within an acceptable response time.

2. Reliability. The system shall ensure consistent detection processing, record storage, alert generation, and dashboard monitoring with minimal errors.

3. Usability. The system shall provide a user-friendly interface for both mobile users and administrators.

4. Security. The system shall protect user accounts, detection records, images, location data, submitted removal records, and ecological monitoring information through authentication, role-based access control, and secure data handling.

5. Compatibility. The system is accessible on mobile devices, with admin functions available within the same Flutter app for users with admin accounts.

6. Maintainability. The system shall allow easy updates, debugging, and improvement of detection, monitoring, reporting, and user management modules.

7. Scalability. The system shall support additional users, reef zones, images, detection records, and monitoring areas without major performance issues.

8. Availability. The system shall be available to authorized users during field monitoring and administrative review.

9. Data Integrity. The system shall maintain accurate, complete, and consistent detection records, GPS data, zone information, alerts, reports, submitted removal records, and ecological monitoring information.

Use Case Diagram

The primary users of the proposed system, as shown in Figure 6, are the Diver, Expert, and Administrator, each performing distinct responsibilities to support the monitoring and management of Crown-of-Thorns Starfish (COTS) infestations. The Diver interacts with the mobile application by logging into the system, uploading underwater images captured using a GoPro camera, and submitting monitoring records for AI-based COTS detection. After the detection process, the Diver can view active COTS records, access the GIS map to monitor affected reef zones, report zone status, and submit removal evidence after intervention activities. The Diver may also add field notes to provide supplementary ecological observations that support reef monitoring and conservation efforts. The Expert serves as the validation authority of the system. Before monitoring records become official, the Expert reviews AI-generated detection results, validates detections with low-confidence predictions, verifies submitted removal evidence, and reviews detection records to ensure the accuracy and reliability of monitoring data.

This validation process improves the credibility of the generated monitoring records before they are used for mapping, analytics, and decision-making. The Administrator interacts with the web-based dashboard to oversee the overall operation of the system. The Administrator can monitor reef zones, view the GIS map, review active COTS records, manage detection alerts, update zone status, generate and export reports, configure system settings, and manage user accounts. The Administrator also monitors validated monitoring records and system analytics to support conservation planning and decision making.

The proposed system is really good for monitoring COTS infestations. It uses intelligence to find problems and experts check to make sure everything is correct. The system also uses maps to show where the problems are and it keeps track of what's happening. It even checks to see if the problems have been fixed and makes reports. This helps people who take care of the reefs to do their job at the time. The system puts all of these things together in one place. It is a complete solution, for dealing with COTS infestations and taking care of the reefs. COTS infestations are a deal and the proposed system is a great way to monitor them.

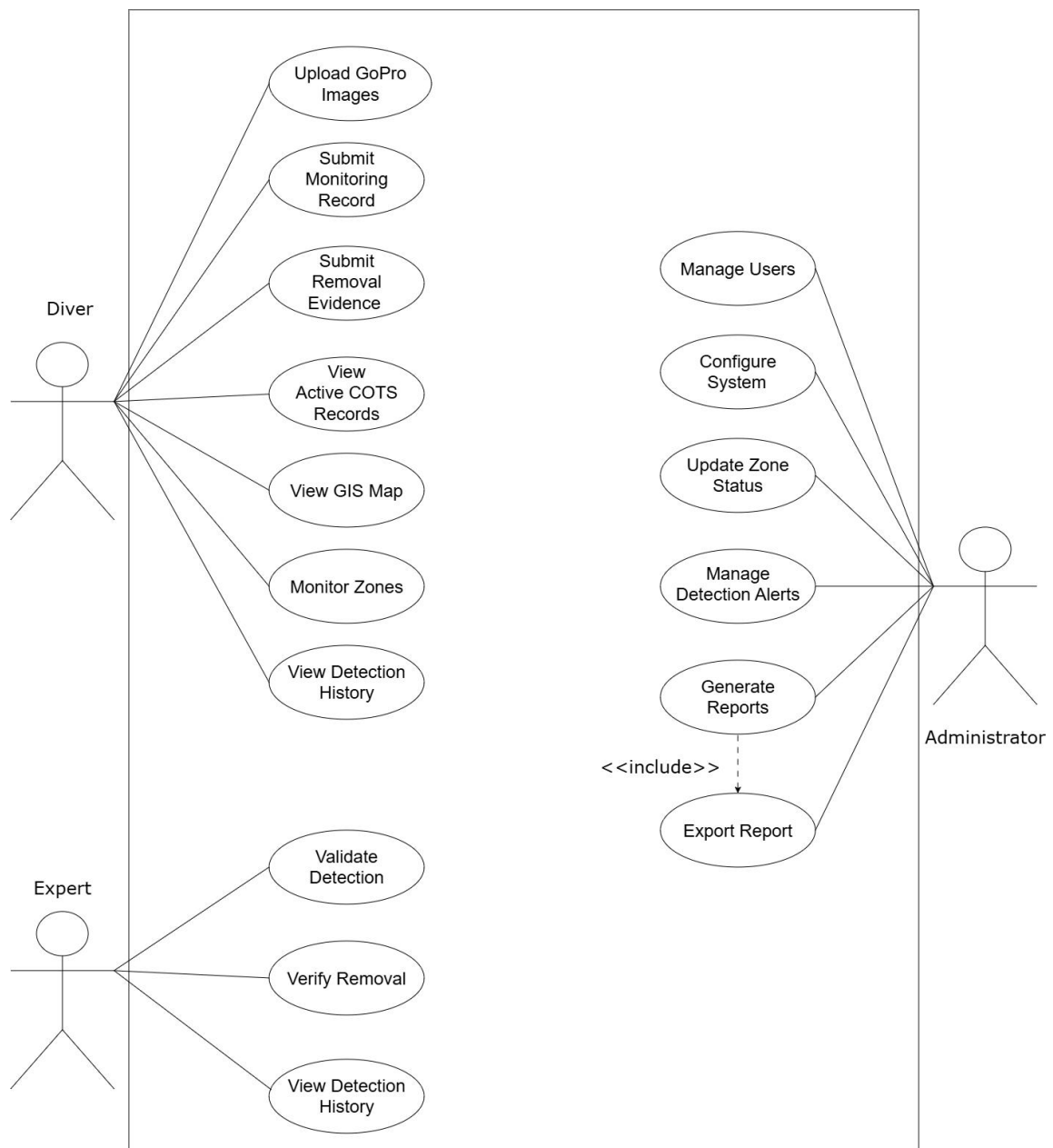


Figure 11. Use Case Diagram of the System

Context Flow Diagram

The Context Flow Diagram (CFD) shown in Figure 7 presents a high-level representation of the proposed COTS Monitor: Crown-of-Thorns Starfish Detection and GIS Mapping System. It illustrates the interactions between the external entities and the system by showing the major data inputs and outputs without describing the internal processing logic. The diagram provides an overview of how monitoring information flows among users and the system to support COTS detection, validation, mapping, intervention, and reporting.

The primary external entity is the Diver, who uses a GoPro camera to capture underwater images and the mobile application to access the system. The Diver submits login credentials for authentication and uploads GoPro images together with location data and monitoring records. Following intervention activities, the Diver also submits removal evidence to document completed COTS removal operations. In return the system gives detection results a GIS map showing the area the status of the detection and the status of the removal. This helps the Diver keep track of the areas that are affected and see how the work to fix things is going.

The Expert is the person who checks and makes sure everything is correct, in the system. The Expert gets the results that the AI found and the proof that the removal was done so they can look at it and decide if it is correct. After reviewing the information, the Expert submits detection validation and removal verification results to ensure that only accurate and verified monitoring records are stored and used for mapping, reporting, and decision-making. This validation process improves the reliability and credibility

of the monitoring data. The Administrator interacts with the system through the web-based dashboard to oversee monitoring operations and system management. The Administrator manages user accounts, updates monitoring zones, configures system settings, and performs monitoring record verification. In return, the system provides reports, exported reports, alert notifications, and analytical summaries that support monitoring activities, conservation planning, and administrative decision-making.

Overall, the Context Flow Diagram illustrates the major exchanges of information within the proposed system, including login data, GoPro images, location data, monitoring records, removal evidence, detection results, GIS map information, validation results, reports, alerts, analytics, and system management data. These data flows demonstrate how field monitoring information is collected, validated, processed, and transformed into meaningful outputs that support efficient Crown-of-Thorns Starfish detection, GIS-based reef monitoring, intervention tracking, and data-driven marine conservation.

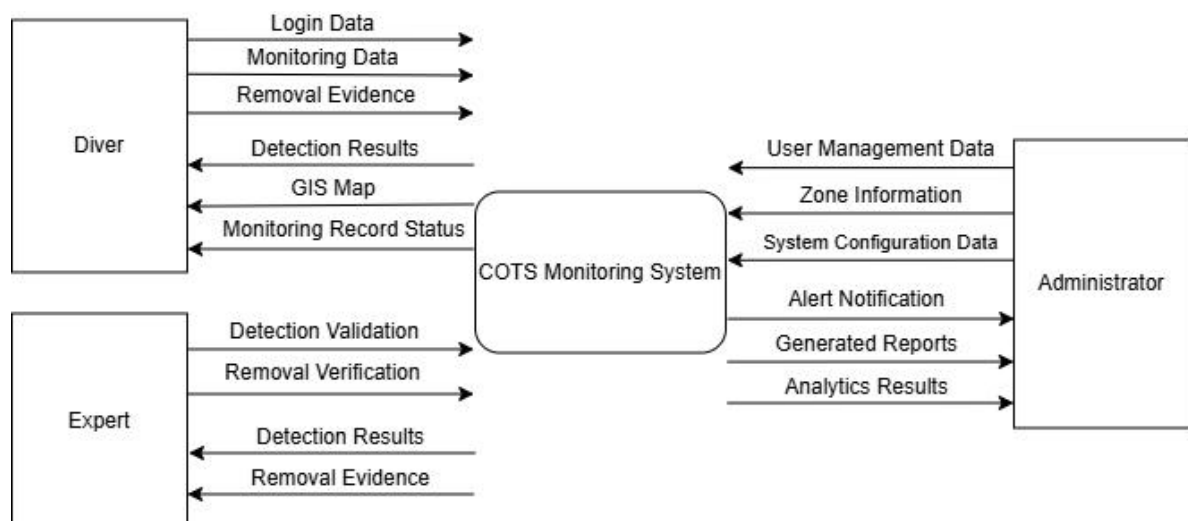


Figure 12. Context Flow Diagram of the System

Data Flow Diagram

The Level 1 Data Flow Diagram (DFD) shown in Figure 8 presents a detailed representation of the proposed COTS Monitor: Crown-of-Thorns Starfish Detection and GIS Mapping System by expanding the processes identified in the Context Flow Diagram. The diagram illustrates how data move between the external entities, system processes, and the centralized database while maintaining consistency with the overall system architecture. The primary external entities interacting with the system are the Diver, Expert, and Administrator.

The process begins when the Diver submits login credentials through the 1.0 Login process. The system authenticates the user's credentials by retrieving the necessary account information from the D1 COTS Monitoring Database before granting access to the system. Once authenticated, the Diver proceeds to the 2.0 Upload Monitoring Data process, where one or more GoPro-captured underwater images, GPS coordinates, reef zone information, date, time, and other monitoring information are submitted. The system supports batch uploading, allowing multiple images collected during a single monitoring activity to be uploaded simultaneously while the Diver is within the target reef zone to ensure accurate location mapping. In accordance with the system design, these uploaded data are first processed by the system and are not immediately stored in the database.

The uploaded monitoring data are forwarded to the 3.0 Preprocess Image process, where image enhancement techniques are applied to improve image quality under underwater environmental conditions. The preprocessed images are then transferred to the 4.0 Detect COTS process, where the

machine learning model analyzes the images to detect and count Crown-of-Thorns Starfish (COTS). The detection results are then evaluated based on their confidence scores. Detections with confidence scores ranging from 80% to 100% are automatically accepted by the system and stored as verified detection records in the D1 COTS Monitoring Database. For automatically accepted or expert-validated detection, the system associate the recorded GPS coordinates with the detection record and also prepares the data for GIS visualization. Conversely, detections with confidence scores below 80% are forwarded to the 5.0 Validate Detection process, where the Expert reviews and verifies the detection results before they are accepted as official monitoring records and stored in the database.

Unlike previous versions of the system, the validation process is performed by the Expert, who reviews the AI-generated detection results before they are accepted as official monitoring records. Only validated detection records are stored in the D1 COTS Monitoring Database, ensuring that subsequent system processes utilize verified monitoring information.

The validated detection records are then processed by the 6.0 Classify Infestation Level module. Using the validated detection results together with the infestation classification framework adapted from the population assessment concepts of Moran and De'ath (1992) and the monitoring and management recommendations of the Great Barrier Reef Marine Park Authority (GBRMPA), the system classifies each monitored reef zone into Low, Moderate, or High infestation levels. These classification results are stored in the database and serve as the basis for subsequent GIS visualization, alert generation, monitoring analytics, and recommended intervention measures.

The infestation classification serves as a decision-support mechanism for prioritizing monitoring activities and identifying reef zones that require management attention.

After classification, the 7.0 Update GIS Map process retrieves the processed detection data from the database and updates the geographic visualization of monitored reef zones. The updated GIS map displays validated COTS locations as red markers based on the recorded GPS coordinates together with their corresponding infestation status. These markers enable Divers and Administrators to immediately visualize newly detected COTS locations and monitor affected reef zones more effectively.

The 8.0 Generate Alerts process retrieves infestation level data from the database to determine whether a monitored reef zone requires immediate attention. When Moderate or High-risk infestations are identified, the system automatically generates alert notifications and displays recommended intervention measures. The system recommends timely removal activities by authorized field personnel for reef zones classified as Moderate or High infestation levels.

Following the intervention, updated monitoring data and removal evidence are submitted for expert verification, while periodic follow-up monitoring is conducted to assess reef recovery and detect any recurring infestations. The generated notifications are sent to the Administrator to support monitoring, response planning, and conservation management. Following field intervention activities, the Diver proceeds to the 9.0 Submit Removal Evidence process by uploading photographs and supporting documentation showing that the detected COTS have been removed. The

submitted evidence is then forwarded to the 10.0 Verify Removal process, where the Expert reviews and verifies the submitted removal evidence. Once verified, the system updates the existing monitoring record in the database by changing its status to indicate that the infestation has been cleared, thereby preserving the complete monitoring history rather than deleting previous detection records.

Finally, the 11.0 Generate Reports and Analytics process retrieves validated monitoring records, GIS information, infestation classifications, and verified removal records from the database to generate reports, monitoring summaries, detection statistics, GIS-based analytics, infestation trends, and intervention summaries. These outputs are provided to the Administrator to support conservation planning, outbreak monitoring, resource allocation, and data-driven decision-making. The generated reports also include recommended intervention measures and the current intervention status of affected reef zones to support conservation planning and management.

The D1 COTS Monitoring Database serves as the centralized repository of the system. It stores user account information, monitoring captures, validated detection records, infestation classifications, GIS monitoring data, verified removal records, alerts, reports, analytics, and audit information through the Created By and Updated By fields to ensure data traceability and integrity.

Overall, the Level 1 DFD illustrates how multiple GoPro-captured underwater images are processed through image preprocessing, AI-based detection, confidence-based validation, infestation level classification, GIS visualization using red location markers, alert generation, removal

documentation, and report generation. This integrated workflow supports accurate COTS monitoring, intervention tracking, and data-driven conservation decision-making. generation. This integrated workflow supports accurate monitoring, effective intervention tracking, and informed decision-making for the management and conservation of coral reef ecosystems affected by Crown-of-Thorns Starfish infestations.

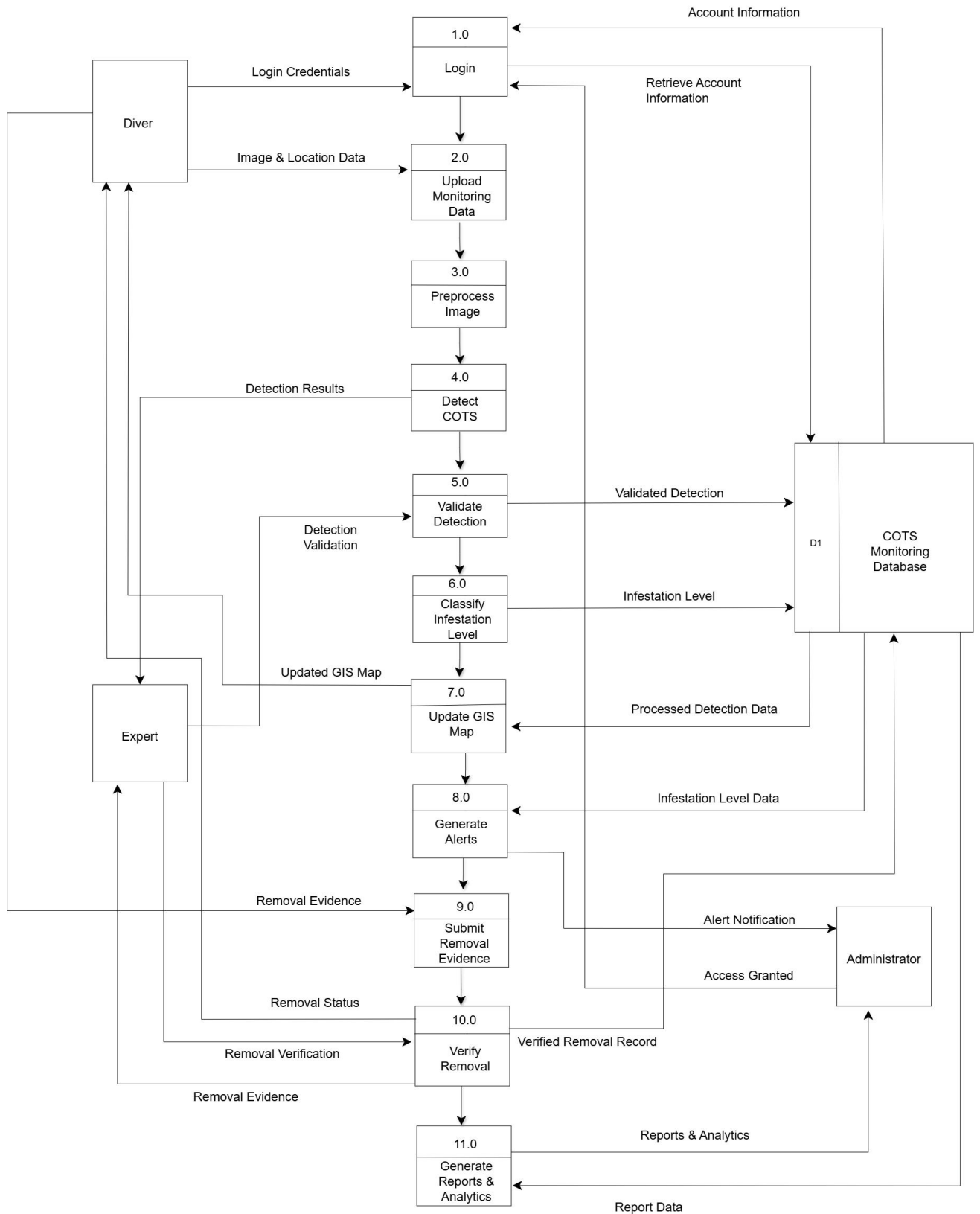


Figure 13. Data Flow Diagram of the System.

SYSTEM DESIGN

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrated in Figure 9 presents the database structure of the proposed COTS Monitor: Machine Learning-Based Crownof-Thorns Starfish Detection and GIS Mapping System. It illustrates how the system's data entities are organized and interconnected through primary and foreign key relationships to support image-based detection, GIS mapping, monitoring, validation, reporting, and conservation management. The database is designed to ensure data integrity, consistency, and efficient retrieval of information throughout the monitoring process.

The Users thing has all the information about people who are registered to use the system. This includes people like Divers and Experts and the people who are in charge which are called Administrators. Each person has their record and that record has things, like their user name and password what they are allowed to do if their account is working or not and when things were done to their record. The entity also serves as a reference for tracking system activities through the created by, updated by, validated by, verified by, and generated by fields found in related tables, ensuring accountability and traceability of user actions.

The Monitoring Capture entity serves as the central repository for field monitoring activities. Each monitoring capture is associated with a specific user and reef zone and stores the uploaded underwater image together with its corresponding geographic coordinates, capture date, monitoring status, and audit information. These records represent the primary data collected

during reef monitoring and serve as the input for the machine learning detection process.

The Reef Zone entity stores information about the monitored reef areas, including the zone name, current infestation risk level, operational status, and audit fields. Each reef zone may contain multiple monitoring captures, allowing the system to organize monitoring records geographically and support zone-based analysis and management. The Detection Record entity stores the results generated by the machine learning model after processing each monitoring capture. Each detection record is directly associated with one monitoring capture and includes the detected COTS count, confidence score, infestation risk level, validation status, validation information, and audit details. Detection records are validated by an Expert before becoming part of the official monitoring database, ensuring the reliability and accuracy of detection results.

The Alert entity is where we keep track of notifications that come from detection records that have been checked and confirmed. Alerts have an important details: the alert level, a message that describes what is happening the status of the system and the time it was created. These notifications help people in charge stay on top of things by letting them know when there are problems or risky conditions, in the areas of the reef that we are watching. We use the Alert entity for these notifications so that people can take action quickly when something serious is detected in the zones we are monitoring.

The Removal Record entity documents intervention activities conducted after COTS detections have been verified. Each removal record is linked to a specific detection record and the user who performed the removal

activity. This thing stores pictures of evidence the dates when things were removed and any comments that were made. It also stores information, about who checked the evidence and when they did it. The entity keeps a record of everything that has happened so we can see what was done in the past. This means the system can keep an eye on things over time and track how well the reef is being fixed.

The Reports entity stores information related to generated system reports, including report type, generation date, file location, and audit information. Reports are generated by authorized users and summarize monitoring activities, validated detections, infestation trends, removal operations, and other analytical outputs that support decision-making and documentation.

The System Settings entity maintains configurable system parameters used by the application, such as operational settings and configuration values. Each change is connected to the administrator who made the update making sure that the system settings are handled correctly.

Links between the parts of the system are created using primary keys and foreign keys to keep the data consistent and make it easy to get the information needed. One user may create multiple monitoring captures, generate multiple reports, update system settings, and submit multiple removal records. A single reef zone may contain multiple monitoring captures. Each monitoring capture produces one corresponding detection record, while a single detection record may generate multiple alerts and multiple removal records to preserve the history of intervention activities. These relationships provide a structured representation of the monitoring workflow, beginning with

field data collection, followed by automated detection, expert validation, alert generation, intervention documentation, and report generation.

Overall, the ERD establishes a comprehensive and normalized database design that integrates machine learning-based COTS detection, GIS-based monitoring, expert validation, conservation intervention tracking, alert generation, and analytical reporting. The database structure is really good for managing data. It helps keep the data consistent and easy to track. This means people can be held accountable for the data. The database structure gives us a base, for watching over marine ecosystems and managing reefs. It also helps us make decisions based on the data we have. The database structure supports ecosystem monitoring and reef management and it helps with evidence-based decision-making.

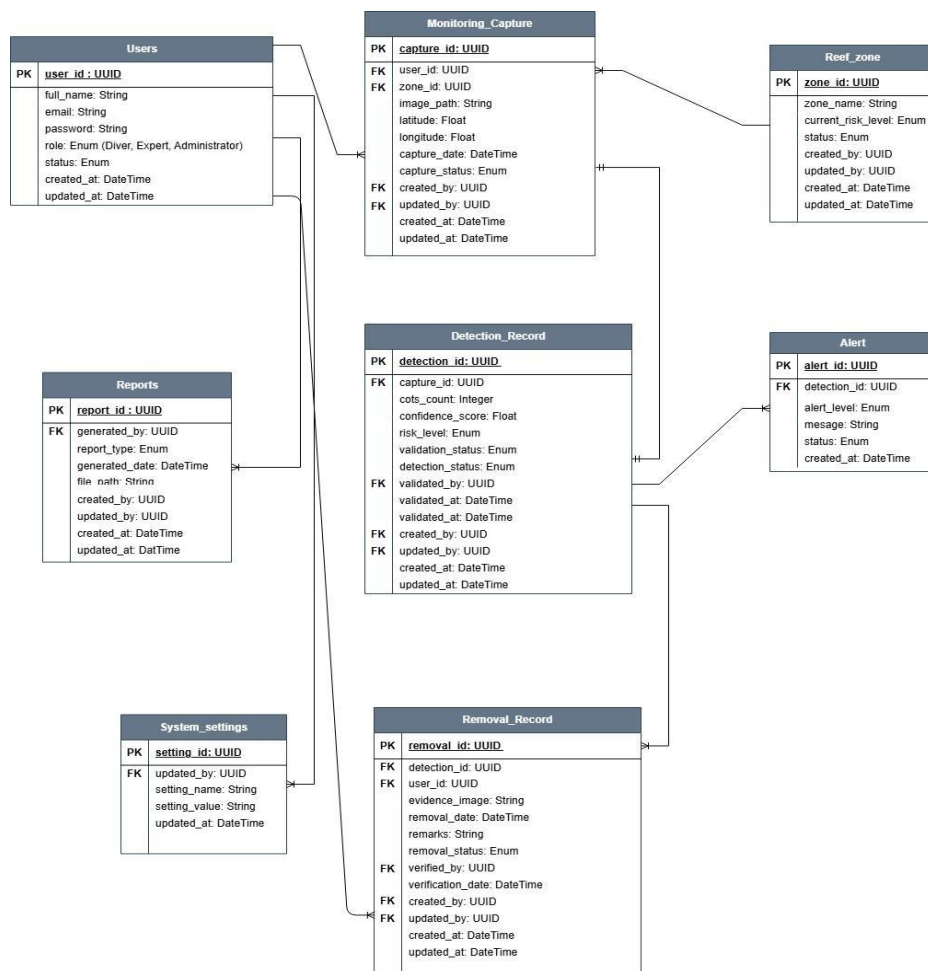


Figure 14: Entity relationship diagram of the system.

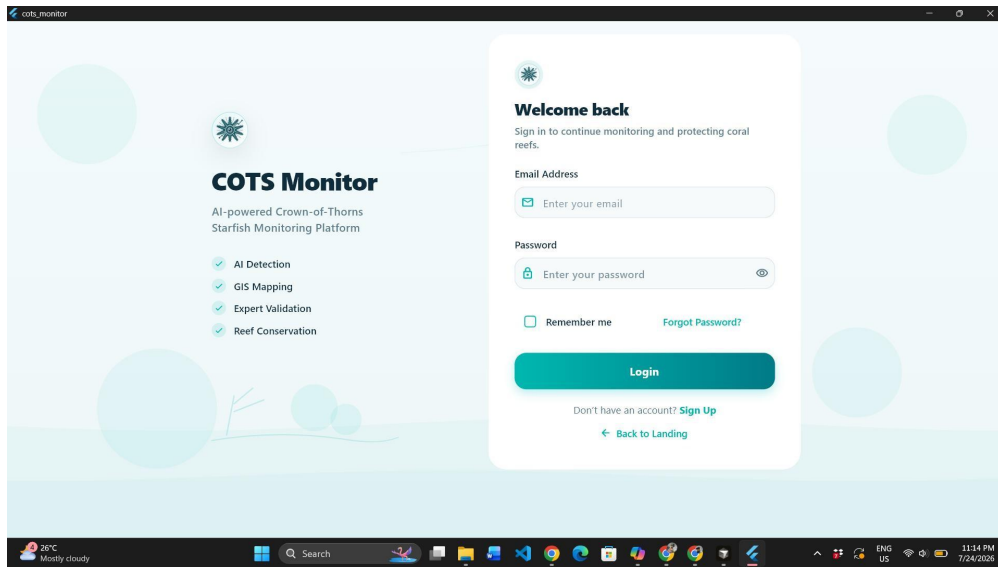
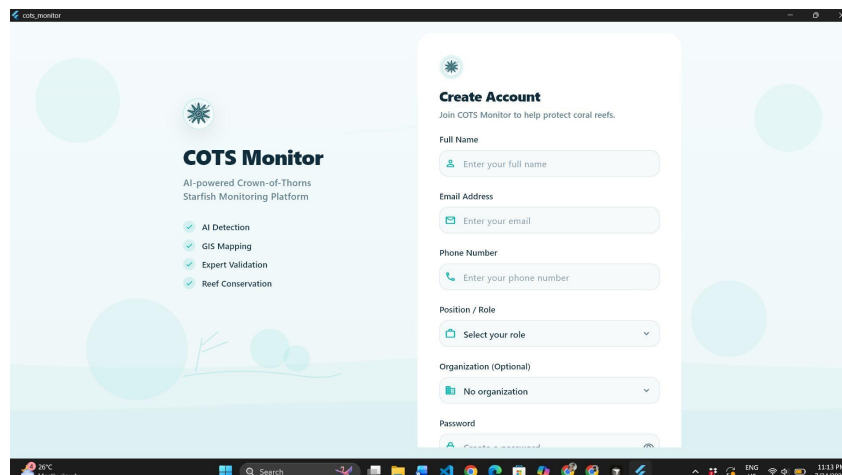


Figure 15. Login Page

Figure 15 presents the Login Page of the proposed COTS Monitor: Machine The Crown-of-Thorns Starfish Detection and GIS Mapping System provides a login page for all the registered users to access the system using their email address and password. It also include useful features such as a Show Password, Remember Me, and Forgot Password to make the logging in easier. After the successful login, the system directs the users to the appropriate dashboard based on their assigned roles as an Administrator, Expert, or Diver. This will help protects the system and to ensures that each user can only use the features available for their role.



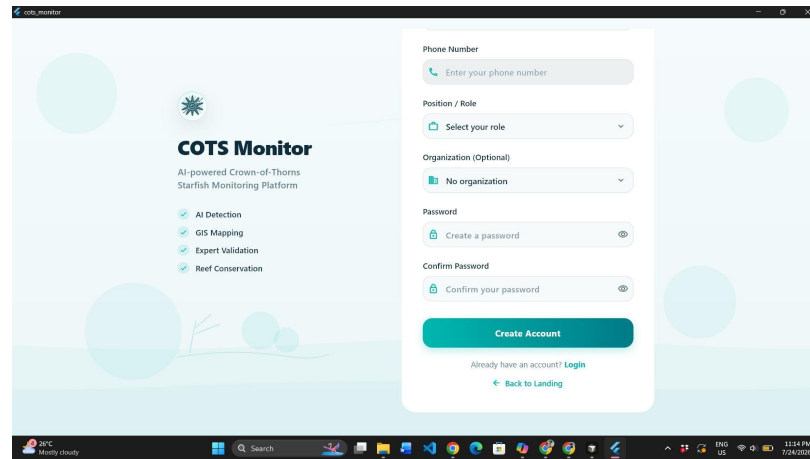


Figure 16. Registration Page

Figure 16 presents the Registration Page allows the new users to create an account by filling in the required personal informations. And during registration, users will provide the details that are needed to create their accounts and verify their identity. Once their account is approved, they are assigned a user role that determines the features they can access in the system. This ensures that the only authorized users can use the system.

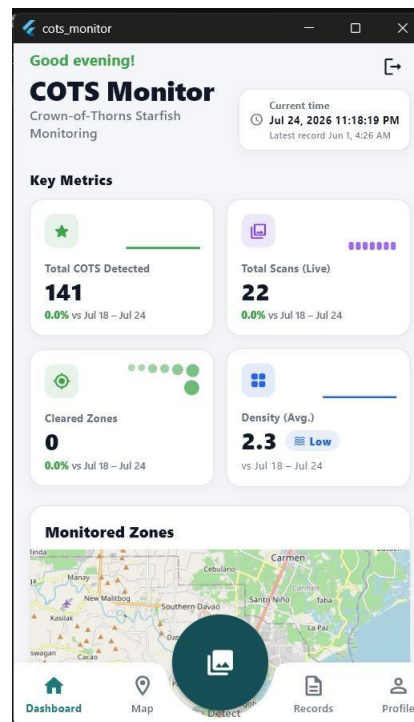


Figure 17. Diver Dashboard

Figure 17 illustrates The Diver Dashboard is the main page that divers see after logging into the system. It gives them an overview of their monitoring activities and also allow them to upload monitoring data, to view their submitted records, to access the GIS map, and check the status of their submissions. This will helps the divers to manage their monitoring tasks more easily and stay updated on their submitted reports.

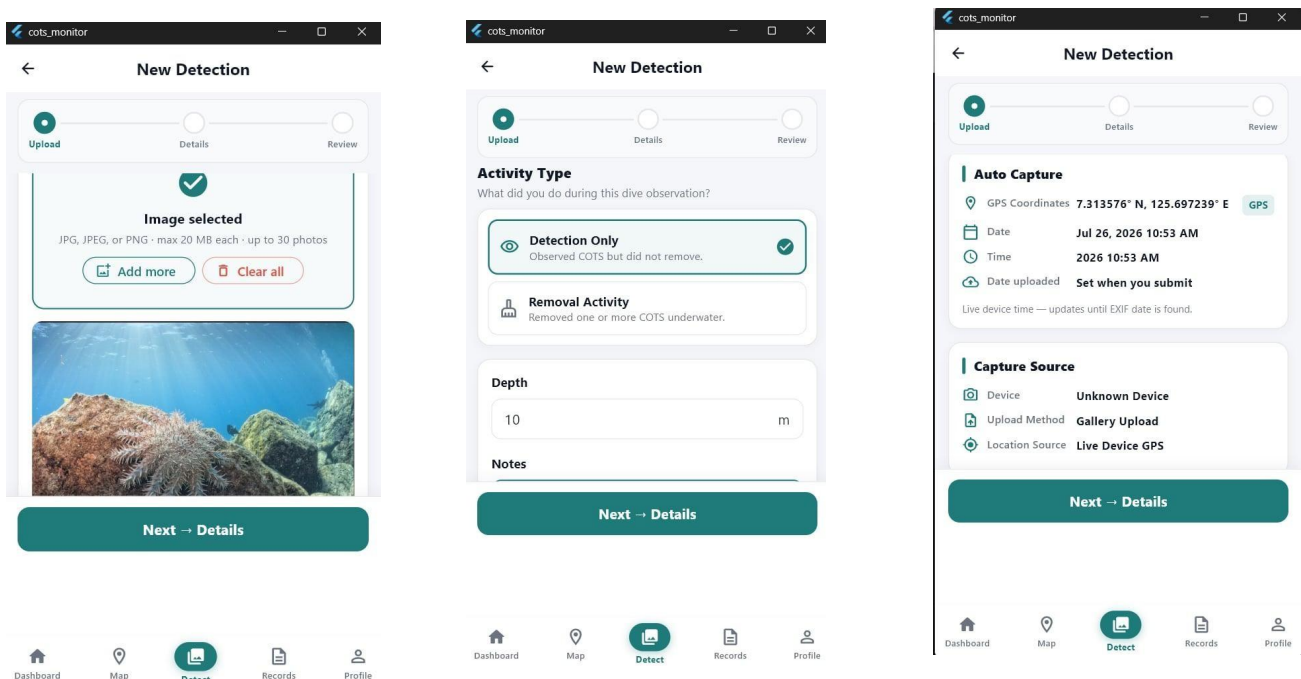


Figure 18. Upload Monitoring Data Page

Figure 18 shows the Upload Monitoring Data page, its where the divers submit a GoPro-captured underwater images together with the GPS coordinates, reef zone information, and other monitoring details. And these data serves as the input for the machine learning pipeline, which performs image preprocessing, object detection, and infestation level classification.

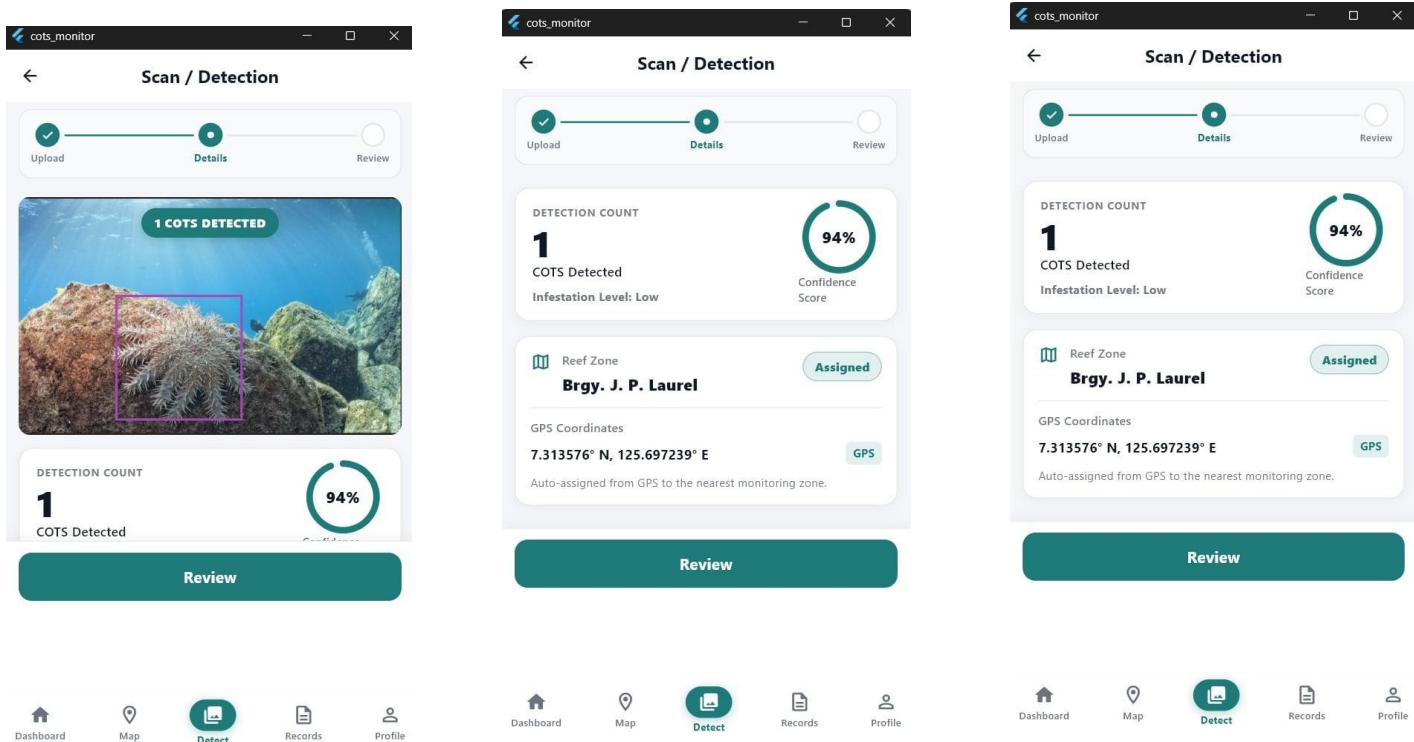
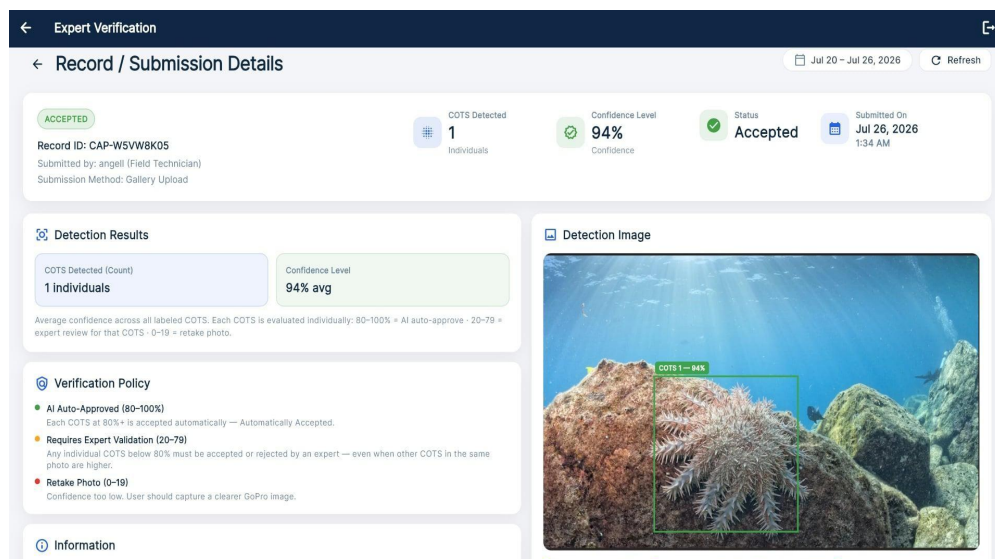


Figure 19. Detection Results Page

Figure 19 presents the Detection Results page, which displays the output generated by the machine learning model after processing uploaded underwater images. The page shows the detected Crown-of-Thorns Starfish (COTS), confidence scores, detection count, and the corresponding infestation level. The results are then forwarded for expert validation.



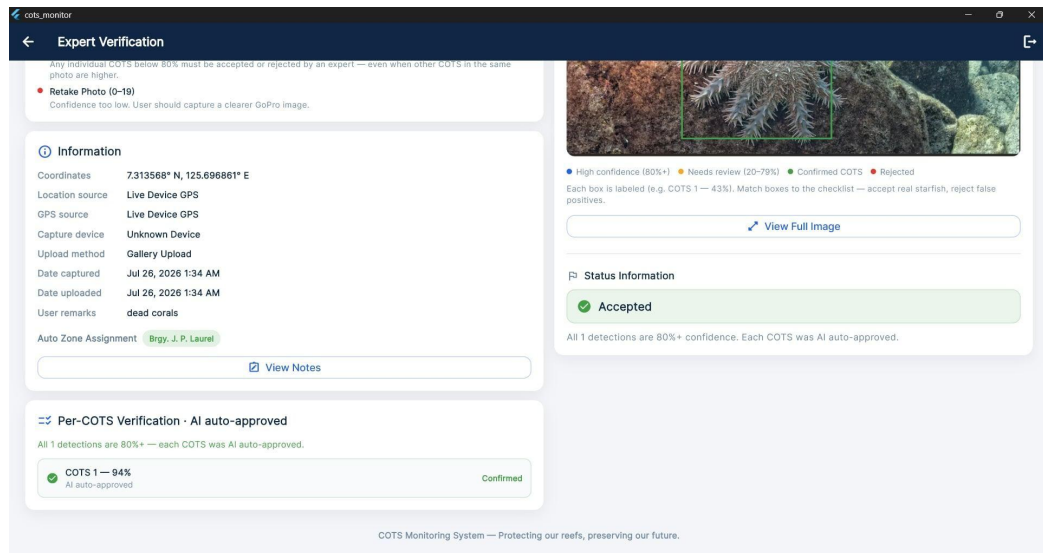


Figure 20. Expert Validation Page

Figure 20 illustrates the Expert Validation page, where detection results with their corresponding confidence scores are reviewed by the Expert. The Expert may approve or reject the AI-generated detections before they are stored as validated monitoring records, ensuring the reliability and accuracy of the monitoring database.



Figure 21. GIS Mapping Page

Figure 21 shows the GIS Map Page shows the reef monitoring zones where the Crown-of-Thorns Starfish (COTS) monitoring activities are carried out. Each colored areas on the map represents specific reef zone, making it easier for all the users to identify different monitoring locations. The map also helps the users to view zone boundaries, see where the COTS have been detected, and better understands the distribution of infestations. This supports monitoring activities and helps in the planning conservation and management efforts.

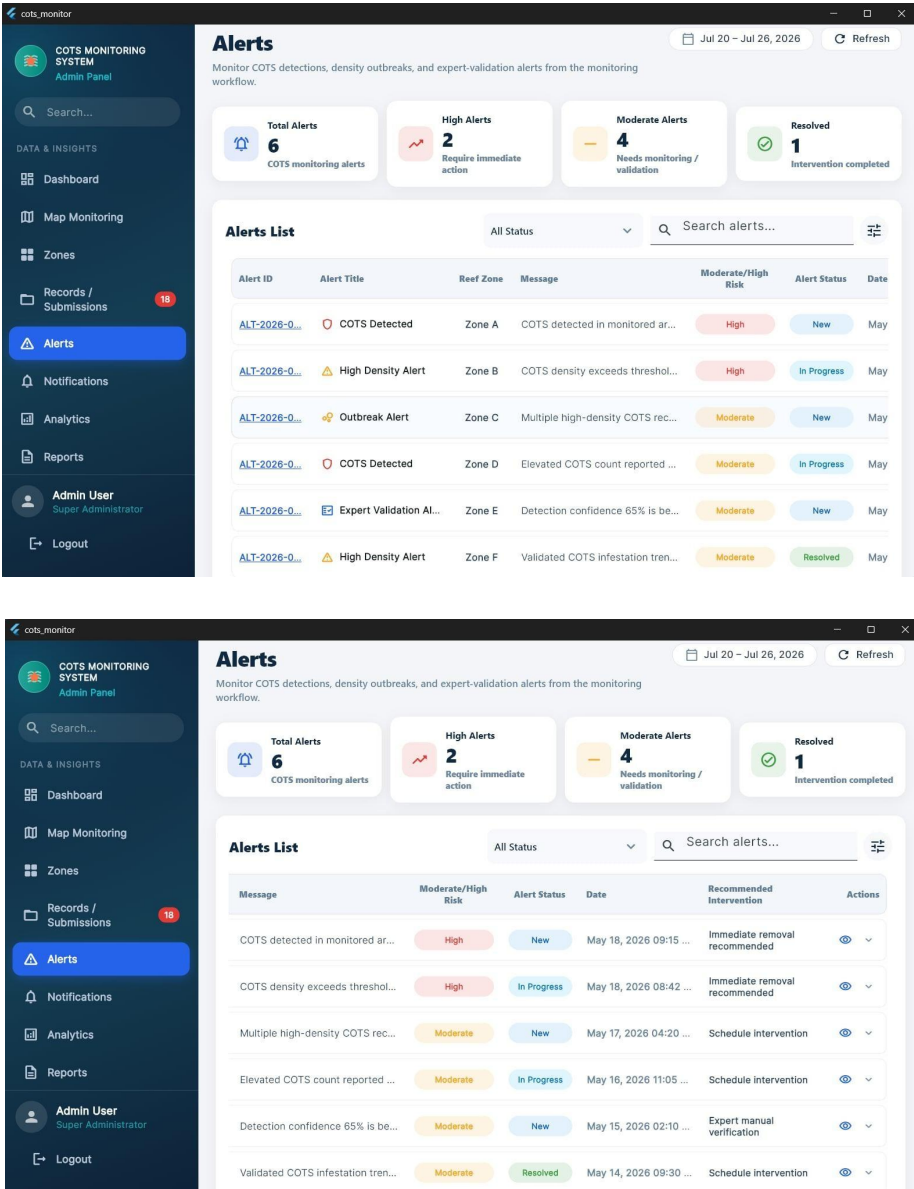


Figure 22. Alert Notification Page

Figure 22 presents the Alert Notification page, which displays notifications for reef zones classified with Moderate or High infestation levels. The page also presents recommended intervention measures to assist administrators in planning appropriate management and conservation activities.

The figure displays five screenshots of the 'Review' page in the 'cots_monitor' application, arranged in two rows. Each screenshot shows a progress bar at the top with three steps: 'Upload' (checked), 'Details' (checked), and 'Review' (active). The bottom of each screen features a navigation bar with icons for Dashboard, Map, Detect, Records, and Profile.

- Top Left Screenshot:** Shows 'Detection Status' as 'Automatically Accepted' with a 'Confidence Score' of 94%. It notes 'The AI detection confidence is 94%. This detection has been Automatically Accepted (80%–100% threshold).' and 'Infestation Level: Low'. Below is a photo of a reef with a label '1 COTS DETECTED' and a 'Submit' button.
- Top Middle Screenshot:** Shows 'DETECTION COUNT' as '1 COTS Detected' and a 'Confidence Score' of 94%. It lists the 'Reef Zone' as 'Brgy. J. P. Laurel' (Assigned) and 'GPS Coordinates' as '7.313573° N, 125.697074° E'. A 'Submit' button is at the bottom.
- Top Right Screenshot:** Shows 'Auto Capture' details: 'GPS Coordinates' (7.313573° N, 125.697074° E), 'Date' (Jul 26, 2026 11:52 AM), 'Time' (2026 11:52 AM), and 'Date uploaded' (Jul 26, 2026 11:52 AM). It also shows 'Capture Source' as 'Unknown Device'. A 'Submit' button is at the bottom.
- Bottom Left Screenshot:** Shows 'Submission Details' including 'Activity Type' (Removal Activity), 'Image File' (sample3.png), 'Depth' (10.0 m), and 'Remarks' (cleared). A 'Submit' button is at the bottom.
- Bottom Right Screenshot:** Shows 'Remarks' (cleared), 'Date Removed' (Jul 26, 2026 11:52 AM), 'Removed Quantity' (1 COTS), and 'Removal Method' (Injection Method). A 'Submit' button is at the bottom.

Figure 23. Removal Evidence Page

Figure 23 illustrates the Removal Evidence Submission page, where divers upload photographs and supporting information after completing COTS removal activities. These submissions serve as documentary evidence for subsequent expert verification and monitoring record updates.

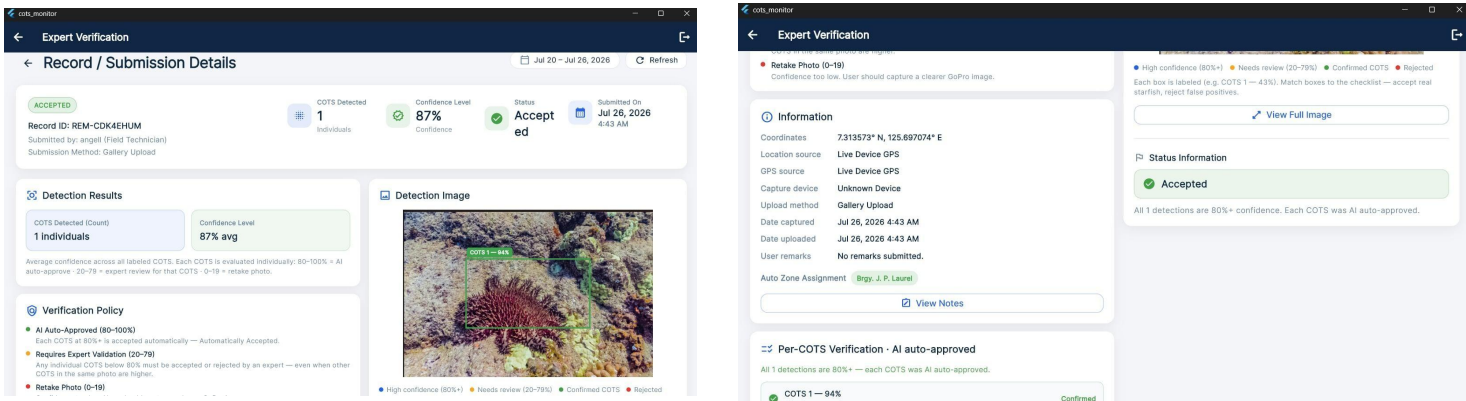


Figure 25. Removal Verification Page

Figure 25 shows the Removal Verification page, where the Expert reviews the submitted removal evidence to verify whether the reported intervention has been successfully completed. Once verified, the monitoring record is updated while preserving the historical record of the detected infestation.

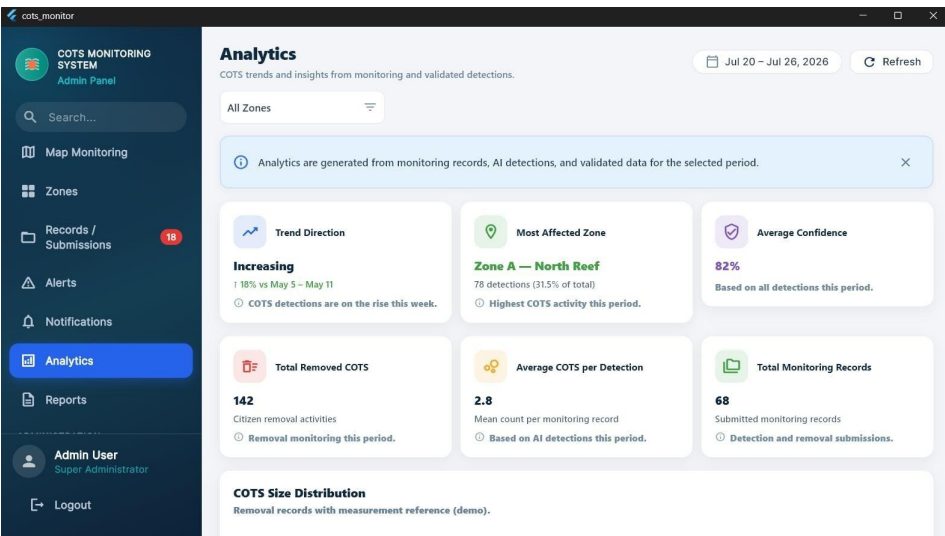


Figure 26. Reports and Analytics Page

Figure 26 presents the Analytics page provides insights into validated COTS monitoring records, AI detection results, and reef zone activities. It summarizes key metrics, detection trends, confidence scores, and monitoring statistics to support effective monitoring and intervention planning. The generated analytics help administrators evaluate COTS distribution and make informed conservation and reef management decisions.

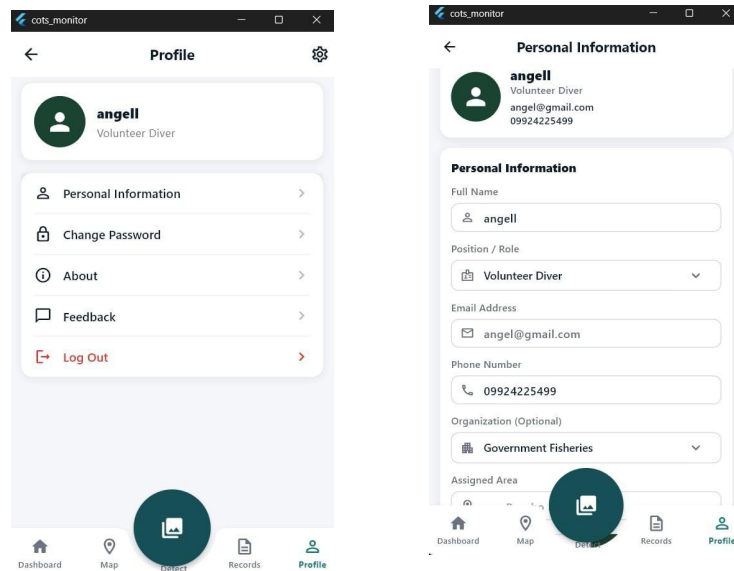


Figure 27. System Settings

Figure 27 presents the System Settings page, where users can manage their account information and personalize system preferences. The interface allows users to update their profile information, change their password, configure notification settings, and perform other account-related functions. These features help maintain account security while improving usability and user experience.

JSON Schema Diagram

The JSON Schema Diagram illustrated in Figure 10 represents the standardized data structures used by the proposed COTS Monitor: Crown-of-Thorns Starfish Detection and GIS Mapping System. It defines how data are organized, validated, and exchanged among the system components, ensuring consistency during storage, processing, and communication between the mobile application, machine learning detection module, GIS mapping module, and web-based administrative dashboard. The diagram shows the plans for the Users Monitoring Capture, Reef Zone, Detection Record, Removal Record, Alert, Reports and System Settings. These plans work together to help the system monitor things check that everything is okay create reports and manage conservation.

The Users schema defines the structure for storing user information, including unique user identifiers, full name, email address, password, assigned role (Diver, Expert, or Administrator), account status, and audit timestamps. This schema provides secure authentication and role-based authorization, allowing users to access only the functions assigned to their roles within the system.

The Monitoring Capture schema represents every monitoring activity submitted by field personnel. Each monitoring record is linked to a user and a reef zone through foreign keys and stores information such as the uploaded GoPro image path, geographic coordinates (latitude and longitude), capture date, monitoring status, and audit information. This schema serves as the

primary source of field monitoring data used for machine learning-based image analysis.

The Reef Zone schema stores information about monitored reef locations, including the zone name, current risk level, monitoring status, and audit fields identifying the users who created or updated the zone information. This schema provides the geographic framework for organizing monitoring records and visualizing infestation levels through the GIS mapping component. The Detection Record schema stores the output generated by the machine learning detection model after processing each monitoring capture. Every detection record is associated with a specific monitoring capture and contains the detected COTS count, confidence score, calculated infestation risk level, validation status, validation information, and audit fields. These validated detection records serve as the primary basis for monitoring infestation trends, updating reef conditions, generating alerts, and supporting conservation decision-making.

The Removal Record schema defines the structure for documenting Crown of Thorns Starfish removal activities performed in the field. Each removal record is linked to a validated detection record and the user who performed the removal. It stores supporting evidence images, removal date, remarks, verification status, verifier information, verification date, and audit fields. This plan keeps a record of everything that has been done to deal with problems. It also lets experts check if the things that were supposed to be removed are really gone.

The Alert plan is, about messages that the system sends out when it finds out that there are many pests somewhere Each alert is associated with a specific detection record and stores information such as the alert level, notification message, alert status, and creation timestamp. These alerts support timely monitoring and response by notifying administrators whenever critical reef conditions require immediate attention.

The Reports schema defines the structure for system-generated reports. Each report records the report type, generation date, output file location, and the administrator who generated the report, together with audit information. These reports summarize monitoring activities, detection results, verified removal records, GIS updates, and infestation statistics to support conservation planning, management decisions, and documentation.

The System Settings are where we keep all the system parameters that the administrator is in charge of. The System Settings have things, like the name of the setting what the setting is set to who made the change and when they made it. This way the System Settings help us keep track of how the system's set up and who made those changes to the System Settings.

Each schema has rules that say what attributes are needed. It also includes data types like UUID, String, Integer, Float, DateTime and Enum. It also has key and foreign key relationships. These help keep the data correct and consistent, in the database. These validation mechanisms ensure that monitoring records, detection outputs, removal evidence, alerts, reports, and configuration settings conform to the required data structure before being stored or processed by the system.

Furthermore, the standardized JSON schemas promote interoperability among the mobile application, machine learning detection module, GIS mapping services, and administrative dashboard by providing a consistent format for data exchange through application programming interfaces (APIs). This simple system helps keep the data consistent and easy to manage. It also makes the system work better and last longer. The Crown-of-Thorns Starfish monitoring is more efficient with this system. It helps with validation and conservation reporting, for the Crown-of-Thorns Starfish. We can also use it for reef management activities related to the Crown-of-Thorns Starfish. This makes everything run smoothly and reliably for the Crown-of-Thorns Starfish projects.

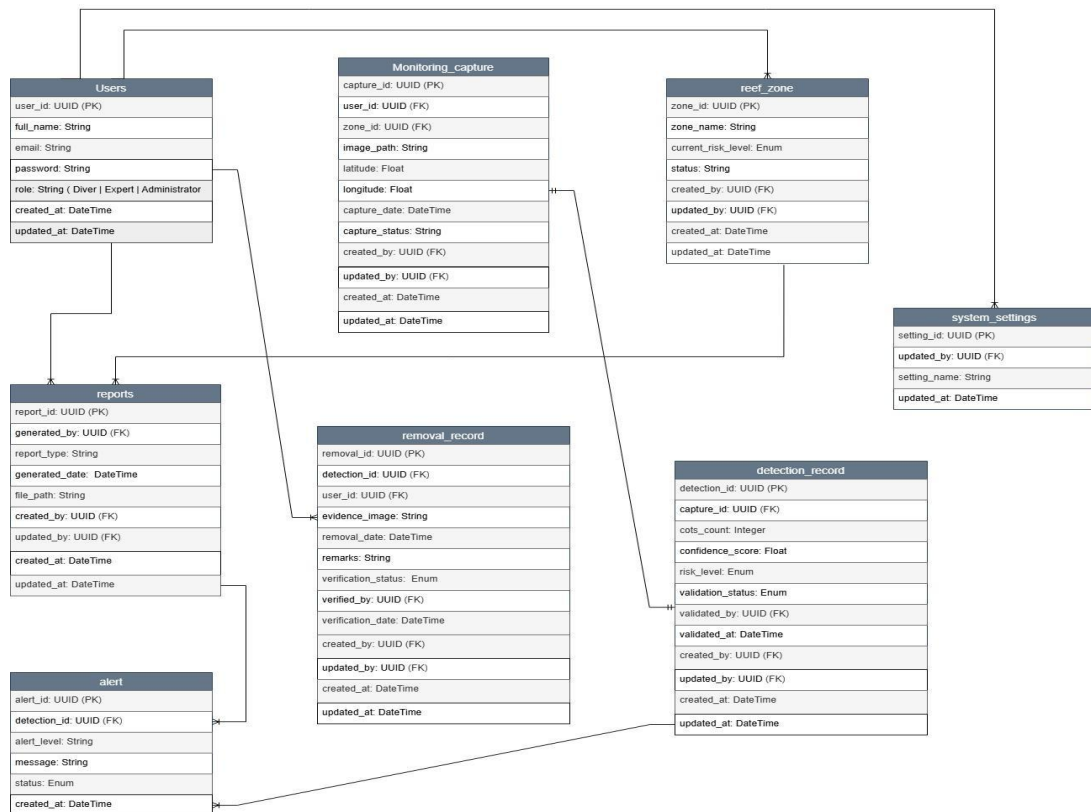


Figure 28: JSON Schema Diagram

Data Dictionary

The Data Dictionary presents a detailed description of all data elements used in the proposed system, serving as a reference for the system's database structure and data organization. It defines the tables, fields, data types, and descriptions to ensure consistency, accuracy, and proper understanding of how data is stored and managed within the system. This section is essential in guiding the implementation of the database and supporting system development, maintenance, and future enhancements.

Table 2. Data Dictionary

Table Number	Name of the Table
2.1	users
2.2	monitoring_capture
2.3	reef_zone
2.4	detection_record
2.5	removal_record
2.6	alert
2.7	reports
2.8	system_settings

users			
Field Name	Data Type	Constraints	Description
user_id	INT(200)	PK, NOT NULL, AUTO INC.	Unique identifier for each account

username	VARCHAR(100)	NOT NULL	Login username of the user
email	VARCHAR(150)	NOT NULL, UNIQUE	Email address for registration & notifications
password	VARCHAR(255)	NOT NULL	Encrypted user password.
role	ENUM('Diver','Expert','Administrator')	NOT NULL	User role in the system.
full_name	VARCHAR(200)	NOT NULL	Full name of the registered user
status	ENUM('Active','Inactive')	NOT NULL	Account status: active or inactive

Table 2.1: users

The Users table stores the information of all authorized individuals who can access the COTS Monitor system, including Divers, Experts, and Administrators. It contains user credentials, assigned roles, account status, and audit timestamps to support secure authentication and role-based access control. The table also serves as a reference for monitoring captures, validation activities, report generation, and other system transactions, ensuring accountability and traceability of user actions.

monitoring_capture			
Field Name	Data Type	Constraints	Description
capture_id	INT	PK, NOT NULL, AUTO INC.	Unique identifier for each uploaded image

user_id	INT	FK → users NOT NULL	ID of the officer/diver who submitted it
zone_id	INT	FK → reef_zones NULL	ID of reef zone where image was captured
image_path	VARCHAR(500)	NOT NULL	File path of the uploaded GoPro image.
latitude	DECIMAL(10,8)	NOT NULL	Latitude of capture location.
longitude	DECIMAL(11,8)	NOT NULL	Longitude of capture location.

Table 2.2: monitoring_capture

The Monitoring_Capture table stores all monitoring activities submitted by field personnel. It records the uploaded GoPro image, associated reef zone, geographic coordinates, capture date, and processing status. Each monitoring record is linked to a specific user and reef zone, serving as the primary source of data for machine learning-based COTS detection and GIS mapping.

reef_zone			
Field Name	Data Type	Constraints	Description
zone_id	INT(11)	PK, NOT NULL, AUTO INC.	Unique reef zone identifier.
zone_name	VARCHAR(100)	NOT NULL	Detection record that triggered the alert
current_risk_level	ENUM('Low','Mode	NOT NULL	Current
	rate','High')		infestation level of the reef zone.

status	ENUM('Active','Inactive')	NOT NULL	Current monitoring status.
created_by	UUID	FK, NOT NULL	User who created the reef zone record.
updated_by	UUID	FK, NULL	User who last updated the reef zone record.
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Date and time when the reef zone record was created.
updated_at	DATETIME	NULL	Date and time when the reef zone record was last updated.

Table 2.3: reef_zone

The Reef_Zone table contains information about the monitored reef areas within the system. It stores the reef zone name, current infestation risk level, monitoring status, and audit information indicating who created or last updated the record. This table supports zone-based monitoring and allows the system to organize detection records geographically for easier visualization and management.

detection_record			
Field Name	Data Type	Constraints	Description
detection_id	UUID	PK, NOT NULL	Unique identifier of the detection record.
capture_id	UUID	FK, NOT NULL	Monitoring capture associated with the detection.

cots_count	INT	NOT NULL	Number of detected Crown-of-Thorns Starfish.
confidence_score	DECIMAL(5,2)	NOT NULL	Detection confidence score.
risk_level	ENUM('Low','Moderate','High')	NOT NULL	Calculated infestation risk level.
validation_status	ENUM('Pending','Validated','Rejected')	NOT NULL	Validation status of the detection.
validated_by	UUID	FK, NULL	Expert who validated the detection.
validated_at	DATETIME	NULL	Date and time of validation.
created_by	UUID	FK, NOT NULL	User who created the detection record.
updated_by	UUID	FK, NULL	User who last updated the record.
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Creation timestamp.
updated_at	DATETIME	NULL	Last update timestamp.

Table 2.4: detection_record

The Detection_Record table stores the results generated by the machine learning model after processing uploaded underwater images. It records the detected COTS count, confidence score, calculated infestation

level, validation status, and expert validation details. Each detection record is linked to a monitoring capture and serves as the basis for GIS mapping, alert generation, intervention planning, and reporting.

removal_record			
Field Name	Data Type	Constraints	Description
removal_id	INT	PK, NOT NULL, AUTO INC.	Unique identifier for each reef zonereport
detection_id	INT	NOT NULL	Descriptive name of the reef zone
user_id	INT	NULL	Narrative of the zone's geographic location
evidence_image	ENUM	NOT NULL	GPS latitude of the reef zone center
removal_date	ENUM	NOT NULL	GPS longitude of the reef zone center
remarks	TEXT	NOT NULL DEF: active	Status: active, scheduled, or cleared
verification_status	DATETIME	NULL	Total area of reef zone in hectares
verified_by	DATETIME	NOT NULL DEF: NOW()	Timestamp when the zone was added
verification_date	DATETIME	NULL	Timestamp of the last update to zone

created_by	UUID	FK, NOT NULL	User who created the removal record.
updated_by	UUID	FK, NULL	User who last updated the record.
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMES	Creation timestamp.

Table 2.5: removal_record

The Removal_Record table documents intervention activities performed after confirmed COTS detections. It stores removal evidence, removal date, remarks, verification status, and expert verification details. By maintaining these records, the system preserves a complete history of removal activities while supporting monitoring, validation, and conservation management without deleting previous records.

alert			
Field Name	Data Type	Constraints	Description
alert_id	UUID	PK, NOT NULL	Unique identifier of the alert.
detection_id	UUID	FK, NOT NULL	Detection record associated with the alert.
alert_level	ENUM('Low','Mode rate','High')	NOT NULL	Severity level of the generated alert.
message	TEXT	NOT NULL	System-generated notification message.
status	ENUM('Unread','Read','Resolved')	NOT NULL	Current alert status.

created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMES TAMP	Date and time the alert was generated.
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Table 2.6: alert

The Alert table stores notifications automatically generated when validated detection records indicate significant COTS infestations. Each alert contains the severity level, notification message, current status, and corresponding detection record. This table enables the Administrator to monitor high-risk reef zones and initiate appropriate intervention planning.

reports			
Field Name	Data Type	Constraints	Description
report_id	UUID	PK, NOT NULL	Unique identifier of the report.
generated_by	UUID	FK, NOT NULL	Administrator who generated the report.
report_type	ENUM('Daily','Weekly','Monthly','Custom')	NOT NULL	Type of generated report.
generated_date	DATETIME	NOT NULL	Date and time the report was generated.
file_path	VARCHAR(255)	NOT NULL	Storage location of the generated report.
created_by	UUID	FK, NOT NULL	User who created the report record.
updated_by	UUID	FK, NULL	User who last updated the report record.

created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMES TAMP	Record creation timestamp.
updated_at	DATETIME	NULL	Last update timestamp.

Table 2.7: reports

The Reports table stores information about reports generated by the system for monitoring and documentation purposes. It includes the report type, generation date, file location, and the administrator responsible for generating the report. The table supports the production of tabular records, monthly reports, analytics, and other summary reports required for monitoring and decision-making.

system_settings			
Field Name	Data Type	Constraints	Description
setting_id	UUID	PK, NOT NULL	Unique identifier for each system setting
updated_by	UUID	FK, NOT NULL	Administrator who modified the setting.
setting_name	VARCHAR(100)	NOT NULL	Name of the configurable system parameter.
setting_value	VARCHAR(255)	NOT NULL	Value assigned to the system parameter.
updated_at	DATETIME	NOT NULL	Date and time the setting was last modified.

Table 2.8: system_settings

The `System_Settings` table stores configurable system parameters managed by the Administrator. It contains the setting name, assigned value, and information about the user who last modified the configuration. This table allows administrators to maintain and update system configurations while preserving audit records of configuration changes.

Technologies, Concepts, and Theories

This section presents the key models, algorithms, and emerging technologies utilized in the development of the Machine Learning-Based Crown-of-Thorns Starfish (COTS) Detection and GIS Mapping System. It explains how data is processed from acquisition to decision-making, as well as the tools and technologies used to implement the system. The discussion follows the major stages of the system workflow, including data collection, pre-processing, feature extraction, model training, and classification.

Data Collection

Data collection is performed using underwater imaging devices, primarily GoPro cameras equipped are utilized for field image acquisition, as described in the system architecture. Field Officers or Divers capture images of coral reef environments during monitoring activities. Each image is accompanied by metadata such as GPS coordinates, date, time, and reef zone information. The collected images are transmitted through the mobile application and stored in the centralized database. These datasets are annotated using bounding boxes to label Crown-of-thorns starfish (COTS), serving as ground truth for training and validating the machine learning model [6].

Annotation Tools

To support accurate machine learning model training, image annotation tools are used to label Crown-of-Thorns Starfish (COTS) within collected underwater images. In this study, Roboflow was used to annotate the image dataset using bounding box techniques. The dataset consists of 70% locally collected underwater images captured using a GoPro camera during reef monitoring activities and 30% publicly available COTS images obtained from online datasets. Each image was manually annotated by drawing bounding boxes around visible COTS instances to generate the ground truth labels required for supervised object detection. The combination of locally collected and publicly available images improves dataset diversity while ensuring that the model is trained using images representative of the local marine environment. Proper annotation enhances model learning, improves detection accuracy, and increases the robustness of the machine learning pipeline under varying underwater conditions.

Data Pre -Processing

Before analysis, the collected underwater images undergo pre-processing to improve quality and reliability. Underwater images are often affected by poor visibility, light distortion, and water turbidity [8].

Pre-processing techniques include:

- Contrast enhancement
- Color correction
- Noise reduction
- Image resizing and normalization

- Auto-Orient

These steps improve image clarity and ensure consistency, leading to better detection accuracy and reduced false results [20], [21].

Dataset Balancing Strategy

To improve the performance and generalization capability of the machine learning model, a dataset balancing strategy was implemented during dataset preparation. The annotated dataset consists of 70% locally collected underwater images captured using a GoPro camera and 30% publicly available Crown-of-Thorns Starfish (COTS) images. This combination ensures that the model is trained using images representative of the local reef environment while increasing dataset diversity through publicly available samples.

To further address class imbalance and improve model robustness, data augmentation techniques such as image rotation, horizontal and vertical flipping, scaling, cropping, and brightness adjustment were applied to the training dataset. These augmentation methods increase the variability of COTS samples without altering their essential characteristics, enabling the model to learn under different underwater conditions, lighting variations, viewing angles, and image orientations.

The balanced and augmented dataset enhances the model's ability to accurately detect Crown-of-Thorns Starfish while reducing the risk of overfitting and improving its performance across diverse underwater environments.

Feature Extraction

Feature extraction is performed automatically by the convolutional layers integrated within the selected YOLO object detection models (YOLOv8, YOLOv11, and YOLO26). These models learn and extract discriminative visual features from underwater images without the need for manual feature engineering. The extracted features include:

- Shape and structure of Crown-of-Thorns Starfish (COTS)
- Texture patterns
- Color distribution
- Spatial relationships and object boundaries

These visual features enable the models to accurately distinguish COTS from surrounding coral reef environments under varying underwater conditions. The extracted features are then used for object localization, classification, and confidence score prediction, improving the overall accuracy and robustness of the detection system [13], [14].

Model Training Workflow

The model training workflow consists of several stages to develop and evaluate machine learning models for the detection of Crown-of-Thorns Starfish (COTS). Following image annotation and dataset preparation, the dataset is divided into training, validation, and testing subsets using an 80:10:10 ratio. The training dataset is used to train each of the three selected object detection models: YOLOv8, YOLOv11, and YOLO26. The validation dataset is used to optimize hyperparameters such as the learning rate, batch size, and number of training epochs, while the testing dataset is used to

evaluate the final performance of each trained model using previously unseen images.

To determine the most suitable model for the proposed COTS Monitoring System, the performances of YOLOv8, YOLOv11, and YOLO26 are compared using standard object detection evaluation metrics, including Precision, Recall, F1-score, mean Average Precision at 50% Intersection over Union (mAP@50), and mean Average Precision across multiple IoU thresholds (mAP@50–95) [15], [16]. The model that achieves the highest overall detection performance while maintaining reliable accuracy under varying underwater conditions is selected for integration into the proposed system.

This comparative evaluation ensures that the selected model provides the best balance of detection accuracy, precision, and robustness, supporting reliable monitoring and conservation of coral reef ecosystems affected by Crown-of-Thorns Starfish infestations.

Classification

The proposed COTS Monitoring System utilizes deep learning-based object detection models to detect and classify Crown-of-Thorns Starfish (COTS) from underwater images. The study evaluates and compares the performance of YOLOv8, YOLOv11, and YOLO26, which are selected for their capability to perform real-time object detection with high accuracy under varying underwater conditions.

Each model performs the following object detection tasks:

- Object localization using bounding boxes

- Classification of COTS and non-COTS objects
- Confidence score prediction for each detected object

These capabilities enable the automatic identification and counting of Crown-of-Thorns Starfish, making the models suitable for marine ecosystem monitoring and conservation applications.

To determine the most suitable model for the proposed system, YOLOv8, YOLOv11, and YOLO26 are trained and evaluated using the same annotated dataset and training parameters. Their performances are compared using standard object detection evaluation metrics, including Precision, Recall, F1-score, mean Average Precision at 50% Intersection over Union (mAP@50), and mean Average Precision across multiple IoU thresholds (mAP@50–95). The model that demonstrates the best overall performance in terms of detection accuracy, robustness, and reliability under underwater conditions will be selected for integration into the proposed COTS Monitoring System.

Technologies Used in the System

Machine Learning Pipeline

The Machine Learning Pipeline serves as the core processing component of the proposed COTS Monitoring System. It consists of a sequence of processes that transform underwater images into validated detection results through image preprocessing, annotation, model training, object detection, expert validation, and performance evaluation. This structured workflow enables the system to accurately detect and classify Crown-of-Thorns Starfish (COTS) while ensuring reliable monitoring outputs.

The pipeline begins with the collection of underwater images captured using a GoPro camera, supplemented by publicly available COTS images to improve dataset diversity. The collected images undergo preprocessing and manual annotation using bounding boxes to generate labeled datasets for supervised learning. The annotated dataset is then used to train and evaluate the three selected object detection models: YOLOv8, YOLOv11, and YOLO26. Each model learns to identify COTS based on visual characteristics such as shape, texture, and spatial features extracted from underwater images.

Following model training, the models are evaluated using standard object detection metrics, including Precision, Recall, F1-score, mAP@50, and mAP@50–95, to determine the most suitable model for integration into the proposed system. The selected model is then incorporated into the detection module, where it automatically identifies COTS, generates bounding boxes, predicts confidence scores, and estimates the number of detected starfish. After the detection process, the system evaluates the confidence score of each detected Crown-of-Thorns Starfish (COTS). Detections with confidence scores ranging from 80% to 100% are automatically accepted by the system because they satisfy the predefined confidence threshold established for this study.

These accepted detections are stored as verified monitoring records in the database and are subsequently used for GIS visualization, alert generation, reporting, and monitoring activities. Conversely, detections with confidence scores below 80% are forwarded to the Expert for manual validation. The Expert reviews the detection results and determines whether

they should be accepted or rejected before they are stored as official monitoring records. This hybrid validation approach reduces the Expert's workload while maintaining the accuracy and reliability of the monitoring system.

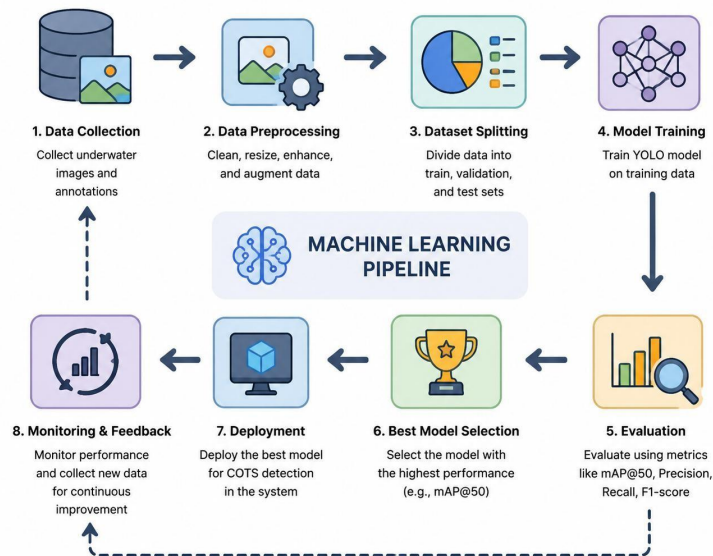


Figure 29. Machine Learning and Computer Vision

GoPro-Based Imaging System

The system uses GoPro cameras as the primary image acquisition device, while the mobile application serves as the upload, monitoring, and GPS synchronization platform. Field users capture underwater images using GoPro cameras mounted for high-resolution underwater image capture. The application also records GPS location, date, time, and reef zone information for each image. These data are transmitted to the server for processing and storage. This approach ensures real-time data collection and supports efficient field monitoring operations.

The use of GoPro cameras was selected because of their superior underwater imaging capability, durability, wide-angle lens, wireless connectivity, and ability to operate at greater depths compared to standard mobile phone cameras. These features improve image quality and increase the accuracy of machine learning-based Crown-of-Thorns Starfish (COTS) detection in reef environments.



Figure 30. GoPro-Based Imaging System

Geographic Information System (GIS)

The proposed system integrates Geographic Information System (GIS) technology to visualize the spatial distribution of validated Crown-of-Thorns Starfish (COTS) detections. GPS coordinates collected during underwater image capture are used to map the locations of detected COTS within monitored reef zones. This enables administrators and marine resource managers to identify affected areas, monitor infestation patterns, and assess the spatial distribution of COTS populations. In addition to spatial visualization,

the GIS module supports intervention planning by highlighting reef zones classified as having Moderate or High infestation levels. These classifications serve as the basis for generating alerts and recommending appropriate intervention measures. Following field intervention, verified removal records and updated monitoring information are reflected on the GIS map, allowing users to monitor the current status of affected reef zones while maintaining historical monitoring records.

The visualization of validated detection records, infestation levels, and verified removal activities supports effective decision-making, resource prioritization, conservation planning, and long-term monitoring of coral reef ecosystems. By integrating machine learning outputs with GIS-based spatial analysis, the system provides a comprehensive platform for monitoring and managing Crown-of-Thorns Starfish outbreaks [22], [23].

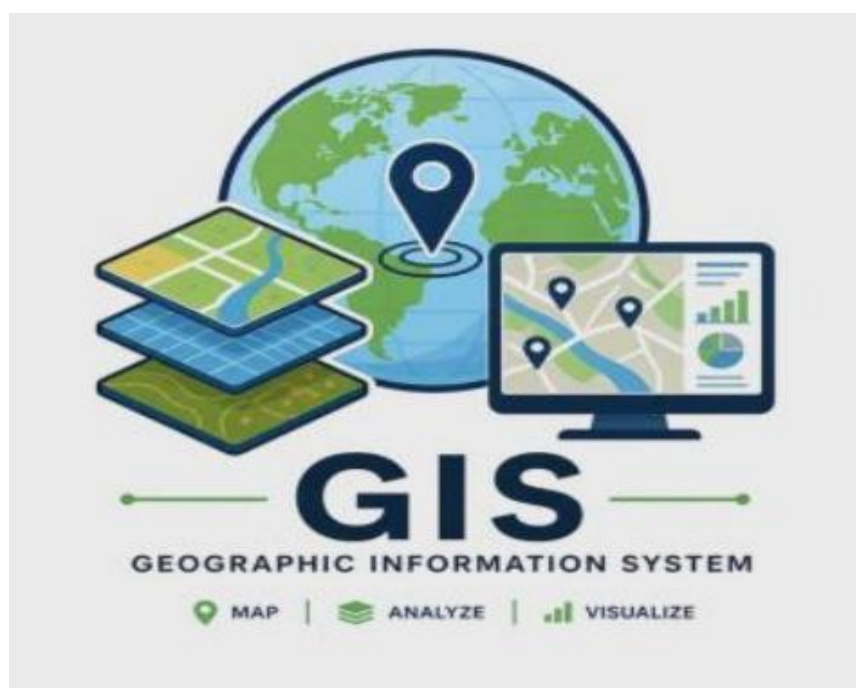


Figure 30. Geographic Information System

Database Management System (DBMS)

The proposed system uses Firebase as its cloud-based Database Management System (DBMS) to store, organize, and manage monitoring information. Firebase serves as the centralized repository for system data, including user accounts, monitoring captures, reef zones, validated detection records, removal records, alerts, reports, analytics, and system settings. The database supports efficient storage, retrieval, and updating of records while maintaining data integrity and consistency across system modules.

In accordance with the system workflow, uploaded monitoring data are first processed through the machine learning pipeline and validated by the Expert before the resulting detection records are stored in the database. Verified removal records and updated monitoring statuses are also maintained to preserve the complete monitoring history rather than deleting existing records. This centralized cloud-based data management supports GIS visualization, alert generation, report generation, analytics, and conservation planning while improving the scalability, reliability, and maintainability of the proposed system.

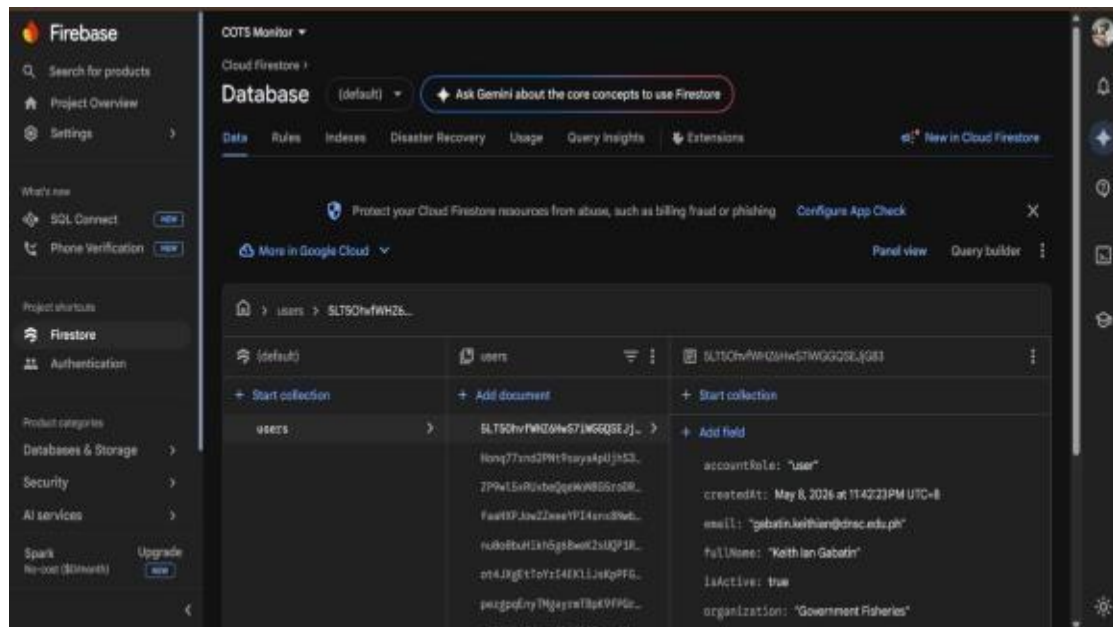


Figure 31. Database Management System

Image Processing Techniques

Image processing techniques are applied during the preprocessing stage of the Machine Learning Pipeline to enhance underwater images before annotation, model training, and object detection. The collected images, including those captured using a GoPro camera, may be affected by underwater conditions such as low visibility, color distortion, uneven lighting, and image noise. To improve image quality, preprocessing techniques such as contrast enhancement, color correction, image resizing, and noise reduction are applied before the images are used for annotation and model training.

These preprocessing techniques improve the visibility of Crown-of-Thorns Starfish (COTS) and enhance the quality of the annotated dataset, enabling the selected object detection models (YOLOv8, YOLOv11, and YOLO26) to learn more representative image features. By providing higher-quality input images, the preprocessing stage contributes to improved

detection accuracy, robustness, and overall model performance under varying underwater environmental conditions [20], [21].



Figure 32. Image Processing Techniques

Hardware Specifications

The proposed system requires adequate hardware resources to support image processing, machine learning model training, system deployment, and field monitoring activities. For model development and evaluation, a computer equipped with at least an Intel Core i7 or AMD Ryzen 7 processor, 16 GB of RAM, and an NVIDIA RTX-series GPU is recommended to efficiently train and compare the three selected object detection models: YOLOv8, YOLOv11, and YOLO26. For field deployment, a GoPro camera is used to capture high-resolution underwater images because of its waterproof design, durability, and suitability for marine environments. A GPS-enabled mobile device is also used to record geographic coordinates and upload monitoring information, including captured images, to the proposed system.

The system utilizes Firebase as its cloud-based database to store user information, monitoring captures, validated detection records, reef zone data, removal records, alerts, reports, and analytics. This centralized cloud infrastructure supports efficient data storage, retrieval, and management while enabling secure access to monitoring information across different system modules. These hardware and cloud computing resources provide the computational capability required for image processing, machine learning model evaluation, GIS visualization, and overall system operation, ensuring reliable performance for marine ecosystem monitoring.

SYSTEM TESTING AND IMPLEMENTATION

The System Testing and Implementation phase ensures that the proposed Machine Learning-Based Crown-of-Thorns Starfish (COTS) Detection and GIS Mapping System functions according to its specified requirements. Testing will be conducted to verify the integration of major system components, including image acquisition, image preprocessing, machine learning-based detection, expert validation, database management, GIS visualization, alert generation, removal record management, and report generation. Functional and non-functional testing will be performed to evaluate system accuracy, reliability, usability, and overall performance.

The machine learning component will be evaluated by comparing YOLOv8, YOLOv11, and YOLOv10 using Precision, Recall, F1-score, mAP@50, and mAP@50–95. The best-performing model will be integrated into the proposed system. The results of the testing process will verify that the system operates accurately and reliably in supporting marine ecosystem

monitoring, COTS detection, GIS-based visualization, and conservation planning.

Train-Test-Validation Split

The collected dataset consists of 865 annotated underwater images of Crown-of-Thorns Starfish (COTS), comprising 605 locally collected images captured using a GoPro camera (70%) and 260 publicly available images obtained from online datasets (30%). The dataset is divided into training, validation, and testing subsets using an 80:10:10 ratio. Approximately 692 images are allocated for training, 87 images for validation, and 86 images for testing. The training dataset is used to train the three selected object detection models—YOLOv8, YOLOv11, and YOLO26. The validation dataset is used to optimize model parameters and minimize overfitting, while the testing dataset is reserved for evaluating the performance of each model using previously unseen images. This data-splitting strategy ensures a reliable and unbiased assessment of model performance.

Model Testing and Performance Metrics

The performance of the three selected object detection models—YOLOv8, YOLOv11, and YOLO26—will be evaluated using standard object detection metrics, including Precision, Recall, F1-score, mean Average Precision at 50% Intersection over Union (mAP@50), and mean Average Precision across multiple Intersection over Union thresholds (mAP@50–95). Precision measures the proportion of correctly identified Crown-of-Thorns Starfish (COTS) detections, while Recall evaluates the

model's ability to detect all relevant COTS instances within the dataset. The F1-score provides a balanced measure of Precision and Recall, whereas mAP@50 and mAP@50–95 assess the overall object detection performance and localization accuracy of each model. These evaluation metrics will be used to compare the performance of the three models and determine the most suitable model for integration into the proposed COTS Monitoring System.

Confusion Matrix

A confusion matrix will be used to provide a detailed analysis of the detection model's classification performance. It will measure true positives, true negatives, false positives, and false negatives produced during model testing. This allows researchers to identify specific classification errors, such as missed detections or incorrect identification of non-COTS objects. The confusion matrix will support deeper evaluation beyond overall accuracy by showing where the model performs well or poorly. This analysis helps improve model reliability and supports future optimization efforts.

Sample Testing Scenarios

Sample testing scenarios will be conducted to simulate actual underwater monitoring conditions. These scenarios include evaluating images captured under varying lighting conditions, water clarity levels, and coral reef environments to assess the performance of YOLOv8, YOLOv11, and YOLO26. Additional tests will evaluate the system's ability to detect partially visible, overlapping, or multiple Crown-of-Thorns Starfish (COTS) within a single image. System functionality will also be tested by verifying GoPro

image uploads, GPS coordinate capture, expert validation of detection results, infestation level classification, GIS map updates, alert generation, removal evidence submission, and report generation. These testing scenarios ensure that both the machine learning models and the integrated system operate effectively under practical deployment conditions.

Benchmark Comparisons

Benchmark comparisons will be conducted to evaluate the performance of the three selected object detection models—YOLOv8, YOLOv11, and YOLO26—for Crown-of-Thorns Starfish (COTS) detection. The models will be compared using standard object detection evaluation metrics, including Precision, Recall, F1-score, mAP@50, mAP@50–95, inference speed, and computational efficiency. The comparative analysis will identify the model that provides the best balance of detection accuracy, robustness, and processing performance under varying underwater conditions. In addition to model comparison, the system's detection results will be compared with manual field observations and expert validation to assess the reliability of the automated detection process. The effectiveness of the integrated workflow—including expert validation, GIS-based mapping, infestation level classification, alert generation, removal record management, and report generation—will also be evaluated to determine whether the proposed system improves monitoring efficiency and supports informed conservation decision-making.

SYSTEM MAINTENANCE

The System Maintenance phase focuses on ensuring the continuous operation, reliability, and improvement of the COTS Monitor: Crown-of-Thorns Starfish Detection and GIS Mapping System after deployment. Maintenance activities include monitoring system performance, identifying and resolving issues, and implementing updates to improve functionality, usability, and adaptability. Regular maintenance will involve updating the mobile application, improving the machine learning models through dataset expansion and retraining using newly collected underwater images, and maintaining the accuracy of GIS visualization and stored monitoring records.

Maintenance also includes preserving the integrity of validated detection records, removal records, reports, and analytics stored in the database. The expert validation workflow and database performance will be regularly reviewed to ensure the consistency and reliability of monitoring information. In addition, completed removal records and infestation trends will be monitored to support follow-up monitoring and assess changes in reef conditions over time. User feedback and system evaluation results will also be used to guide future enhancements, ensuring the long-term efficiency, reliability, and sustainability of the proposed system.

System Security Plan

To safeguard system data and resources, a comprehensive System Security Plan will be established. The system will implement secure login mechanisms, Multi-Factor Authentication (MFA), and role-based access control to ensure that only authorized users, such as Administrators, Experts,

and Divers, can access system functions based on their assigned roles. Data transmitted between the mobile application, machine learning module, GIS module, and cloud database will be protected using secure communication protocols, while stored information—including user accounts, validated detection records, GPS coordinates, removal records, reports, and analytics—will be secured through encryption and regular database backups.

Access to monitoring records and removal records will be controlled through role-based permissions and expert validation mechanisms to prevent unauthorized access, modification, or deletion of system data. Routine security checks, vulnerability assessments, and penetration testing will be conducted to identify potential security risks and strengthen the overall security of the system. An incident response and escalation procedure will also be implemented to ensure that security issues are properly reported, assessed, and resolved according to their severity.

Table 4 presents the System Security Plan based on ISO/IEC 27001 standards, ensuring that the proposed system follows structured and internationally recognized information security practices. These controls help protect sensitive information, maintain data integrity, and support the confidentiality, availability, and reliability of the COTS Monitor: Crown-of-Thorns Starfish Detection and GIS Mapping System.

Security Domain (ISO 27001)	Control Area	Description / Implementation in the System
A.5 Information Security Policies	Policy Implementation	The system will establish structured security rules that govern how data is handled, accessed, and protected throughout system operations
A.6 Organization of Information Security	Roles and Responsibilities	User roles such as Administrator, Field Officer, and Diver will be assigned with clearly defined duties and controlled system permissions.
A.9 Access Control	Access Management	Access to system features will be regulated through role-based permissions to ensure users can only access authorized data and functions.
A.9 Access Control	Authentication Process	The system will require valid login credentials and enforce secure authentication methods to verify user identity.
A.10 Cryptography	Data Protection	Sensitive information, including detection results, user data, and location details, submitted removal records, and ecological monitoring information, will be protected using encryption during data transfer and storage.

A.11 Physical and Environmental security	Hardware Security	Devices such as GoPro cameras, mobile devices, and servers will be secured in controlled environments to prevent physical tampering or unauthorized use.
A.12 Operations Security	Activity Monitoring	System activities will be recorded and monitored to detect irregular actions and support security auditing.
A.12 Operations Security	Data Integrity Control	Validation mechanisms and error-checking processes will be applied to ensure the accuracy and consistency of system data.
A.13 Communications Security	Network Protection	Secure communication channels (e.g., HTTPS, MQTT with encryption) will be used, along with network restrictions to prevent unauthorized access.
A.14 System Acquisition, Development, and Maintenance	Secure Development	Security practices such as input validation, secure coding, and routine updates will be applied throughout system development and maintenance.
A.16 Information Security Incident Management	Incident Handling	The system will include procedures for identifying, reporting, and resolving security issues to minimize system disruption.

A.17 Information Security Aspects of Business Continuity	Backup and Recovery	Regular data backups and recovery strategies will be implemented to ensure data can be restored in case of failure or data loss.
A.18 Compliance	Regulatory Compliance	The system will follow applicable data protection standards and institutional policies to ensure responsible data management.

Table 11: System Security Plan

System Maintenance Plan

In addition, a System Maintenance Plan will be established to guide the continuous support, improvement, and sustainability of the COTS Monitor: Crown-of-Thorns Starfish Detection and GIS Mapping System after deployment. It defines structured procedures for maintaining system performance, resolving issues, and ensuring long-term functionality of detection, mapping, and monitoring features. The plan incorporates preventive, corrective, adaptive, and perfective maintenance to address system stability, detection errors, environmental changes, and performance enhancement. Scheduled activities include system monitoring, updates, database optimization, validation of detection results and validation of submitted field contribution records to maintain efficiency and accuracy.

These activities ensure that the system remains responsive and capable of supporting reliable monitoring of Crown-of-Thorns Starfish populations.

The System Maintenance Plan, as shown in Table 5, ensures the continuous operation, improvement, and sustainability of the system by aligning maintenance activities with ISO/IEC 25010 quality attributes. It includes procedures for troubleshooting, bug fixing, and handling system errors through proper logging and issue tracking mechanisms. All system updates, detection model improvements, and maintenance activities will be properly documented to ensure continuity and ease of future enhancements. A disaster recovery system will also be implemented, including backups, cloud or local redundancy storage, and system restoration procedures to ensure continuity during unexpected failures. The plan also integrates regular backup and recovery processes to protect detection records, images, and GIS data, ensuring system availability during unexpected failures. This approach guarantees that the system remains reliable, efficient, secure, and adaptable over time.

ISO 25010 Attribute	Maintenance Activity	Description	Frequency
Reliability	System Monitoring	Continuous monitoring of system performance, uptime, and detection processes to ensure stable operation	Daily
Reliability	Detection Record Validation	Verifying accuracy of COTS detection results and input data	Monthly

Performance Efficiency	Performance Optimization	Improving system response time and processing speed of detection and GIS mapping	As needed
Performance Efficiency	Database Optimization	Data cleanup, indexing, and backup of detection records and GIS data	Weekly
Usability	Interface	Enhancing dashboard and	Periodically
	Improvement	user interface based on user feedback	
Security	Security Updates	Applying patches, encryption updates, and vulnerability fixes to protect system data	Regularly
Maintainability	Code Refactoring & Debugging	Fixing bugs and improving system structure for easier maintenance	As needed
Maintainability	Documentation Update	Recording system updates, detection model changes, and maintenance activities	After every update
Compatibility	System Updates	Ensuring compatibility with devices, browsers, and network environments	Periodically

Table 12: System Maintenance Plan.

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